

Serialization patterns of comitatives and instrumentals in German clause structure

Tibor Kiss, Alicia Katharina Börner & Simon Masloch
Linguistic Data Science Lab
Ruhr-University Bochum
{tibor.kiss, alicia.boerner, simon.masloch}@rub.de

Abstract

This paper investigates the serialization of two adverbial modifier types – comitatives and instrumentals – within German clause structure. We challenge the notion of fixed base positions for adverbials, as proposed by Frey & Pittner (1998) and Maienborn et al. (2016). Instead, we show that the observed variability in serialization results from the interaction of weighted linearization constraints, which are sensitive to properties of the modifiers, such as animacy and the presence of specific thematic roles. This approach eliminates the necessity for specific linearization rules for adverbials because the rules relevant for arguments apply to them as well. This uniform treatment challenges proposals that either assume predicate-specific base orders for arguments (Haider 2000b; 2017) or propose dedicated functional projections for the introduction of specific arguments (Fanselow 2003).

We present two empirical studies showing that serialization patterns are governed by preferences rather than categorical rules, and that these preferences are modulated by the semantic interpretation of the preposition.

Comitatives and instrumentals are particularly amenable to comparison, not only because they are classified within the same semantic category (Frey & Pittner 1998; Maienborn et al. 2016), but also because prior analyses have neglected crucial differences in their association with verbal arguments, potential restrictions to specific verb classes, and their shared capacity for both affirmative and privative readings. Given the central role of prepositional semantics, we draw on Rissman & Rawlins (2017) for the analysis of instrumentals and provide an explicit semantic account for comitatives, capturing their ability to select distinct referents.

Our account of serialization is formalized within the framework of Maximum Entropy Grammar (Goldwater & Johnson 2003; Wilson 2006; Hayes 2022), utilizing violable and weighted constraints to generate probability distributions over competing candidate structures thereby capturing optionality.

1. Introduction

Most research on German word order assumes a base order for arguments, with variations attributed to scrambling operations. In contrast, the serialization of adverbial phrases remains controversial. A widely held view posits that “*adverbial phrases can be interspersed freely among the arguments of the verb*” (Uszkoreit 1987: 145; see also Neeleman & Reinhart 1998). However, this assumption has been contested by Frey & Pittner (1998), Haider (2000a), and Maienborn et al. (2016). Although these accounts concur that adverbials are not freely ordered, they diverge substantially in their theoretical approaches. Frey & Pittner (1998) propose distinct base positions for different classes of adverbials, determined by configurational constraints. Haider (2000a) adopts a model of free generation, constrained by the semantics of the adverbials – an account that applies only in contexts involving multiple adverbials. Maienborn et al. (2016) distinguish between V-adjuncts, which attach directly to the verbal cluster (ibid., p. 329),

and VP-adjuncts, which attach at the VP level (ibid., pp. 337, 346).¹ All three approaches associate adverbial placement with semantic factors, with Maienborn et al. (2016) offering the most explicit formulation of this connection.

This paper examines the serialization patterns of comitative and instrumental prepositional phrases (PPs), attributing their surface order primarily to semantic properties, rather than to fixed syntactic base positions and scrambling operations. We adopt a model of free generation, constrained by violable linearization constraints applying within the verb's maximal projection. These constraints are sensitive to both formal and semantic features of the relevant elements – such as thematic roles and animacy – and determine their linear order accordingly. In this analysis, thematic information is relevant only insofar as it is encoded in the linearization constraints. We demonstrate that these constraints influence not only the ordering of arguments but also the positional preferences of adverbials, thereby supporting the hypothesis that arguments and adjuncts are subject to uniform linearization mechanisms. Given this framework, there is no need to posit dedicated functional projections, or a base order derived from thematic hierarchies to account for the observed word order patterns. In adopting this approach, we follow Haider (2000a) in rejecting the existence of fixed base positions for adverbials, while maintaining that their individual distribution is nevertheless shaped by systematic serialization constraints. We not only depart from the assumptions of Frey & Pittner (1998) and Maienborn et al. (2016) concerning the existence of base positions but also demonstrate that the semantic distinctions proposed by Maienborn et al. (2016) are not predictive of the serialization behaviour of comitative and instrumental PPs.

Frey & Pittner (1998) classify comitative and instrumental prepositional phrases as members of a unified adverbial category, which they term *event-internal*, and claim that both exhibit identical serialization behaviour. However, experimental data – discussed in Section 3 and in subsequent sections – do not support this generalization. Our focus on comitative and instrumental PPs is motivated not only by their proposed shared classification (Frey & Pittner 1998; Maienborn et al. 2016), but also by a largely overlooked yet theoretically significant property: both of them exist in affirmative and privative variants.² This semantic opposition is systematically encoded by distinct lexical heads, such as *mit/ohne* in German or *with/without* in English. Affirmative comitatives specify the co-participation of an individual or object alongside another participant within the same event. In contrast, affirmative instrumentals indicate the involvement of an object used by an agent in the same event. Privative comitatives express the absence of such co-participation, while privative instrumentals denote the absence of instrumental involvement. We show that comitatives and instrumentals differ not only in their preferred syntactic positions, but that affirmative and privative comitatives also exhibit distinct positional preferences.

Comitatives and instrumentals function as event modifiers and are associated with participants involved in the events. In line with Maienborn & Schäfer (2011: 1412), we refer to these participants as the *target referents* of the respective modifiers. Importantly, the roles of the target referents within the event vary across different adverbial types, and the choice of the target referent can affect their serialization properties. Instrumental prepositional phrases (PPs) consistently select the agent as their target referent, whereas comitative PPs exhibit greater

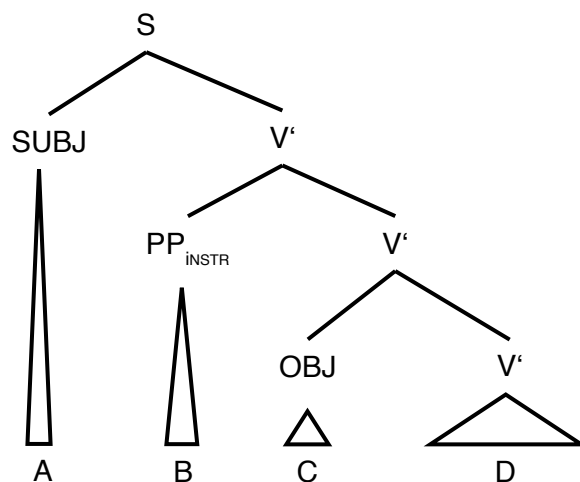
¹ Maienborn et al. (2016) take up the analysis of locatives in Maienborn (2001).

² Schröder's (1986) empirical investigation of German prepositions forms an exception in providing insights into the interpretation of privative comitatives and instrumentals that will be taken up in section 2.2.

flexibility, potentially targeting any argument of the verb. When it is necessary to specify the target referent of a comitative PP, we employ subscripts: *comitative_S* denotes targeting of the subject, while *comitative_O* indicates targeting of the object. Furthermore, comitatives and instrumentals diverge with respect to the verb classes they may modify. Instrumentals obligatorily require the existence of an agent – regardless of its syntactic realization – and are thus confined to action verbs.

Frey & Pittner (1998), along with Haider (2000a) and others, link c-command relations in the German clause to surface order in right-branching binary structures. If a phrase β is minimally c-commanded by a phrase α , it follows that β must immediately follow α . They further argue that comitatives targeting the subject, as well as instrumentals, are required to be minimally c-commanded by the subject in their base position (Frey & Pittner 1998: 511). Consequently, their proposal predicts that both *comitatives_S* and instrumentals must directly follow the subject, as exemplified for an instrumental PP in (1).³

(1) Schematic structure of instrumentals in German clause structure



It is a frequent assumption (Diesing 1992; Frey & Pittner 1998; Haider 2010) that existential *wh*-indefinites are scrambling-invariant. If this is the case, they can act as positional markers.⁴ Given this proviso, the minimal c-command assumption requires that an instrumental adverbial must precede a scrambling-invariant NP object, as is illustrated in (2), because it is base generated there and the *wh*-indefinite cannot scramble across it.

- (2) a. Da hat Hans mittels eines Graphikeditors was erstellt.
there AUX.SG Hans.NOM by.means.of a.GEN graphic.editor.GEN what.ACC create.PTCP

³ As will be taken up in section 4.1, we follow Haider (2010: 58-67) in assuming that there are no functional projections between V and C in German clause structure. Hence, we equate a maximal V projection with the traditional category S.

⁴ Diesing (1992) and Haider (2010) represent the predominant view that existential *wh*-indefinites are invariant under scrambling. In contrast, Lechner (1998: 279) assumes that *wh*-indefinites move out of the VP to be realized within AgrOP. However, this approach encounters difficulties in accounting for remnant topicalization. Struckmeier (2014: 225ff.) follows Lechner's proposal but adds that the potential movement of *wh*-indefinites is semantically motivated, specifically to escape the scope of negation – an issue not relevant to the structures under discussion here. Furthermore, Lechner's analysis is open to criticism in light of Haider's (2010) compelling arguments against the existence of functional projections between V⁰ and C⁰ in the German clause structure.

- b. Da hat Hans was **mittels eines Graphikeditors** erstellt.
 ‘Hans created something using a graphic editor.’

Frey & Pittner (1998) predict that the serialization in (2a) is grammatical, while the one in (2b) is not. However, what we find is that both orders are grammatical, with experimental data indicating a preference for (2b) (see section 3.1). Moreover, instrumental PPs differ from comitative_s PPs in their positional preference, as is illustrated in (3).

- (3) a. Da hat Ida **zusammen mit einem Berater** was unterzeichnet.
 there AUX.SG Ida.NOM together with a.DAT counsellor.DAT what.ACC sign.PTCP
 b. Da hat Ida was **zusammen mit einem Berater** unterzeichnet.
 ‘Ida signed something together with a counsellor.’

In (3), the comitative phrase is preferably positioned to the left of the object. This contrast between examples (2) and (3) stands in opposition to the predictions of Frey and Pittner (1998) and is not addressed in the analyses provided by Haider (2000a) or Maienborn et al. (2016). Haider’s account predicts no positional preference, whereas Maienborn et al. (2016: 329) posit two distinct base positions, each associated with a different interpretation.⁵ While we agree with their assessment regarding locative PPs, we note that this observation does not extend to comitative and instrumental prepositional phrases.⁶ The consideration of comitative_o, as illustrated in (4), introduces additional complexity to the empirical landscape.

⁵ The formulation of Maienborn et al. (2016: 329) is relatively vague when it comes to extending their assumptions about locatives to comitatives and instrumentals: “[Maienborn (2001)] has argued that [the V-adjunction site] can also be targeted by locatives, instrumentals etc. Thus, for instance locatives may be adjoined both at the V-level and at the VP-level.” They continue arguing that comitatives and instrumentals pattern with locatives in this respect, but do not provide evidence for an adjunction site at the VP-level, or at an interpretational difference resulting from attaching comitatives or instrumentals at the VP-level.

⁶ If the two positions signalled a difference in meaning, it should be possible to coordinate a sentence exemplifying one serialization with the negation of a sentence exemplifying the other, thereby yielding a contingent interpretation. For locatives, this can be illustrated with the following example, derived from Maienborn (2001: 195, ex. (11)):

- (i) Angela hat sich im Museum mit Bardo verabredet, aber sie
 Angela.NOM AUX.SG REFL in.the.DAT museum.DAT with Bardo.DAT arrange.to.meet.PTCP but she
 hat sich nicht mit Bardo im Museum verabredet.
 AUX.SG REFL not with Bardo.DAT in.the.DAT museum.DAT arrange.to.meet.PTCP
 ‘In the museum, Angela and Bardo made an appointment, but they didn’t make an appointment to meet in the museum.’

But parallel coordinations of sentences containing comitatives result in a contradiction, as would be expected if both sentences convey the same meaning. In (ii), a *wh*-indefinite cannot be used as the object, since *wh*-indefinites interfere with negation.

- (ii) #Gestern hat Hans ein Handtuch zusammen mit einem Schlafanzug gewaschen
 yesterday AUX.SG Hans.NOM a.ACC towel.ACC together with a.DAT pyjama.DAT wash.PTCP
 aber er hat nicht zusammen mit einem Schlafanzug ein Handtuch gewaschen.
 but he AUX.SG NEG together with a.DAT pyjama.DAT a.ACC towel.ACC wash.PTCP
 ‘Yesterday, Hans washed a towel together with a pyjama, but he did not wash a towel together with a pyjama.’

- (4) a. Da hat Hans **zusammen mit einem Handtuch** was gewaschen.
 there AUX.SG Hans.NOM together with a.DAT towel.DAT what.ACC wash.PTCP
 b. Da hat Hans was **zusammen mit einem Handtuch** gewaschen.
 ‘Hans washed something along with a towel.’

Here, the preference for the serialization (4b) is even stronger than for instrumentals, while Maienborn et al. (2016) would allow both serializations with a difference in interpretation (see fn. 6).

Notably, prior literature, such as Frey & Pittner (1998), Maienborn et al. (2016), as well as almost every other paper concerned with either comitatives or instrumentals (cf. Baker 1997; Zhang 2007; Rissman & Rawlins 2017, among others), discusses only affirmative comitatives and instrumentals. Privative adverbials, as exemplified in (5) and (6), exhibit different behavior. While affirmative and privative instrumentals share ordering preferences, privative comitatives lack such preferences. This optionality, visible in (6), challenges fixed base-position theories. It is unclear why optionality arises only in some contexts if base positions or multiple derivations are assumed.⁷

- (5) a. Da hat Ida **ganz ohne ein Hilfsmittel** was angeschlossen.
 there AUX.SG Ida.NOM entirely without a.ACC means.ACC what.ACC connect.PTCP
 b. Da hat Ida was **ganz ohne ein Hilfsmittel** angeschlossen.
 ‘Ida connected something without using a means.’

- (6) a. Da hat Hans **ganz ohne einen Kollegen** was überprüft.
 there AUX.SG Hans.NOM entirely without a.ACC colleague.ACC what.ACC sift.PTCP
 b. Da hat Hans was **ganz ohne einen Kollegen** überprüft.
 ‘Hans sifted something without a colleague.’

It is a well-known fact that instrumentals always associate with the agent, while comitatives can target all arguments of the modified verb (see Lakoff & Peters 1966; Baker 1997; Zhang 2007, among others). Frey & Pittner (1998) acknowledge this variability, but do not consider the ramifications of two rather different targeting mechanisms for constituent ordering⁸ (whether Maienborn et al. (2016) can adequately account for comitatives_o remains unclear). This distinction yields two primary consequences. First, instrumentals are expected to differ from comitatives in that they may only modify (projections of) action verbs. The obligatory presence of an agent precludes their occurrence with other verb classes, such as non-agentive psychological predicates. Second, the referential flexibility of comitatives introduces potential ambiguity, especially in contexts where both the subject and object are animate. This is exemplified in (7):

- (7) a. Da hat ein Verleger einen Gutachter **zusammen mit einem**
 there AUX.SG a.NOM publisher.NOM a.ACC reviewer.ACC together with a.DAT
 Herausgeber beliefert.
 editor.DAT supply.PTCP

⁷ The same point is addressed in Masloch et al. (2024) concerning optional realizations of the arguments of experiencer-object verbs. See also section 4.2.3.

⁸ This does not only affect comitatives and instrumentals, but also locatives, and attitude adverbials, which are also covered by their minimal c-command condition. Frey & Pittner (1998: 506–511) acknowledge differences in the targeting mechanisms of the *event-internal* adverbials, but this does not lead to modifications in their analysis.

- b. Da hat ein Verleger zusammen mit einem Herausgeber einen Gutachter beliefert.
'A publisher supplied a reviewer together with an editor.'

Example (7a) is ambiguous between a reading where the publisher supplies the reviewer *and* the editor (corresponding to a comitative_O), and a reading where the publisher *and* the editor supplied a reviewer (comitative_S), but targeting the object is preferred. This pattern is reversed in (7b), where a comitative_O interpretation is difficult to obtain. According to Frey & Pittner (1998), neither example can be considered ambiguous: (7a) would only allow comitative_O, and (7b) comitative_S. As far as an application of Maienborn et al. (2016) to the examples in (7) can be envisaged, it appears that they do not predict a difference in preferences but a difference in interpretation. Haider (2000a) would neither predict a difference in preferences nor a difference in interpretation.

In sum, our data indicate that none of the serialization patterns should be judged ungrammatical. Rather, relative preferences and gradient acceptability judgments argue against categorical constraints like minimal c-command. Whether such constraints emerge when multiple adverbials co-occur remains open, but our findings suggest that gradient, violable constraints better capture empirical patterns.⁹

This paper proceeds as follows: Section 2 explores comitative and instrumental PPs, focusing on how each identifies target referents and how affirmative and privative meanings are characterized. Though semantic in origin, these factors directly affect word order. Section 3 reports on experimental results supporting our claims. Section 4 presents the theoretical analysis, using violable, weighted linearization constraints formalized in a Maximum Entropy Grammar (Goldwater & Johnson 2003; Wilson 2006; Hayes 2022). This framework, a probabilistic extension of Optimality Theory (Prince & Smolensky 2004; Müller 2015), allows for variation and preference rather than absolute judgments. Optionality, observed in our data, is incompatible with base-position models but is naturally accommodated by our approach.

2. Properties of Comitative and Instrumental Modifiers

Frey & Pittner (1998) classify comitative and instrumental phrases as members of a single adverbial category, which they term event-internal adverbials. Unlike purely event-modifying adverbials such as temporal modifiers, comitatives and instrumentals do not only modify the event but also establish a relation to a specific target referent. Aside from this peculiarity, event-internal modifiers are assumed to behave analogously to other intersective modifiers, such that predicate modification (cf. Heim & Kratzer 1998) can be employed to combine these modifiers with a verbal projection (see also Maienborn et al. 2016: 323f.).

Frey & Pittner (1998) further posit that comitatives and instrumentals must be minimally c-commanded by their respective target referents. Given the binary right-branching structure of the German verb phrase, this entails that the minimally c-commanding target referent must immediately precede the comitative or instrumental PP, as schematically illustrated in (1). However, this approach fails to adequately capture the distinctive properties of comitatives and instrumentals. Specifically, the target referent for instrumentals is invariably the agent – which may be realized as the subject or as a complement of a prepositional phrase in passive

⁹ Most judgments presented in Frey & Pittner (1998) are graded, indicated by diacritics such as ?, ?? etc. In addition, Frey & Pittner (1998: 492) point out that judgments are subtle, which raises further doubts on a clear-cut categorical analysis.

constructions – whereas the target referent for comitatives is determined more flexibly. Moreover, the proposed parallelism between comitatives and instrumentals is restricted to certain verb classes, namely action verbs, since instrumentals presuppose the presence of an agent, while comitatives do not require a specific thematic role to be present with the modified verb.

An additional dimension has been largely overlooked in previous analyses: comitatives and instrumentals differ not only in the mechanisms by which they identify their target referent and in their combinatorial compatibility with various verb classes, but also with respect to the syntactic implications of the affirmative versus privative distinction. The limited attention afforded to prepositions with a privative meaning has left the syntactic consequences of this distinction insufficiently explored. Although Maienborn (2007, 2019) provides examples of comitative prepositional phrases headed by *ohne* ('without'), these studies do not present evidence for syntactic reflexes corresponding to this semantic contrast, nor is the semantic contrast explained as such.¹⁰ Furthermore, our findings indicate that, while the complement of a comitative may be animate, comitatives do not necessarily require the animacy value of the complement to match that of the target referent. Rather, animacy appears to be determined by the θ -role assigned to the target referent, thereby supporting the claim that the target referent must constitute an argument of the modified verb. Finally, we question the commonly assumed analogy between comitatives and coordinate structures. Coordinated structures allow for interpretations that are unavailable in comitative constructions, and this purported parallelism collapses entirely in the context of privative comitatives.

2.1 (Neglected) Properties of comitatives and instrumentals

It is well-known since the discussion of comitatives in Lakoff & Peters (1966: 120) that the semantic relationship between the complement of the comitative phrase and its target referent is not fixed. Baker (1997: 108) characterizes comitatives as introducing secondary agents and themes, and Zhang (2007: 141) explicitly states that “*comitative constructions show unification of theta-roles*”, from which it follows that the comitative’s “*semantic role is never independent from the role of [the target referent]*”. Instrumental PPs differ from comitatives in that they are fixed to the phrase bearing the *agent* role in the modified event (for a recent analysis, see Rissman & Rawlins 2017: 527–532 and below). The fixation to the *agent* role has immediate consequences for the combinatorial potential of instrumental PPs and differentiates them from comitative PPs. Comitative PPs are not restricted to take *agents* – as in (3) – or *themes* – as in (6) – as target referents. They can also modify the subjects of experiencer-verbs, which are neither agents nor themes, as in (8). Comitatives may also differ in animacy from the target referent, thus contradicting Zhang’s (2007: 136) assumption that the animacy values must be identical, as illustrated in (9).

- (8) Die Kinder mussten zusammen mit ihren Lehrern in
the.NOM.PL child.NOM.PL have.to.PST.PL together with POSS.DAT.PL teacher.DAT.PL in
der Schule frieren.
the.DAT school.DAT freeze.INF
‘The children had to freeze in school along with their teachers.’

¹⁰ Stolz et al. (2006:167-170) mention that comitatives and instrumentals can be (lexically) negated without discussing reflexes of the affirmative/privative distinction.

- (9) Er hat seinen Sohn zusammen mit seinen Sachen abgeholt.
 he AUX.SG POSS.ACC son.ACC together with POSS.DAT things.DAT pick.up.PTCP
 ‘He picked up his son along with his things.’

The verb *abholen* (‘pick up’) in (9) allows animate and inanimate objects alike, which is compliant with differing values of animacy between the object and the complement of the comitative. The options for animacy values for the target referent of a comitative can thus be traced back to the modified verb, to the arguments of the modified verb and the restrictions imposed on that argument.

Maienborn & Schäfer (2011: 1412) argue that “*the identification of these target referents is shown to depend on discourse and world knowledge.*” This assumption is incorrect for instrumentals, which lexically specify that the target referent must be the phrase to which the *agent* role is assigned by the verb (which even holds when this phrase is not syntactically realized, as in passives). In the case of comitatives, referring to discourse or world knowledge cannot account for Zhang’s (2007: 141) assumption that comitatives require unification of θ -roles. Also, discrepancies in the animacy values of the target referent and the complement of the comitative emerge exactly if the modified verb does not impose such a restriction on one of its arguments. This does not only hold for the animacy value of the object, but also carries over to the subject, as will be discussed in section 4.2.3.

Regarding instrumentals, we will follow the analysis in Rissman & Rawlins (2017) and assume that instrumental prepositions modify a verbal projection denoting an event that presupposes an *agent*-relation. The target referent of an instrumental must be the phrase which receives the agent role. In case of suppression (e.g. in passive constructions), the bound between the instrumental and the *agent* role is determined in the semantics and does not require a syntactic reflex. The flexibility of comitatives requires a rather different analysis, as will be discussed in section 2.2.

It has repeatedly been argued (see Lakoff & Peters 1966; Frey & Pittner 1998; Stolz 1998; Zhang 2007, among others) that comitatives are synonymous to a coordination of the target referent of the comitative and the comitative’s internal argument. This is not correct, as can be illustrated with the examples in (10).

- (10) a. Die Kinder haben mit dem Vater ein Rührei
 the.NOM children.NOM AUX.PL with the.DAT father.DAT a.ACC scrambled.egg.ACC
 gekocht. #Die Kinder am Montag, der Vater am Dienstag.
 cook.PTCP. the.NOM children.nom on Monday, the.NOM father.NOM on Tuesday.
 ‘The children cooked scrambled eggs with the father. #The children on Monday, the father on Tuesday.’
 b. Die Kinder und der Vater haben ein Rührei
 the.NOM children.NOM and the.NOM father.NOM AUX.PL a.ACC scrambled.egg.ACC
 gekocht. Die Kinder am Montag, der Vater am Dienstag.
 cook.PTCP. the.NOM children.NOM on Monday, the.NOM father.NOM on Tuesday.
 ‘The children and the father cooked scrambled eggs. The children on Monday, the father on Tuesday.’

Clearly, the interpretation of (10a) differs from that of (10b). Following Stolz et al. (2006: 86, 140), we assume that comitative relations require two entities to be “co-present in the same space (including metaphorical readings of space).” From this perspective, the infelicity of continuing the first sentence of (10a) with the second is unsurprising, whereas the same continuation is entirely acceptable in the coordinated structure of (10b). The crucial difference lies in

the fact that the first sentence in (10a) describes a single event, while the coordination in (10b) is ambiguous, permitting either a single or multiple event interpretation.

Except for Schröder (1986), prior analyses of comitatives and instrumentals ignore that comitative and instrumental relations are not only introduced by *affirmative* prepositions but also by *privative* ones. While the preposition *mit* ('with') introduces an affirmative relation, its antonym *ohne* ('without') provides the corresponding privative relation. Privative comitatives or instrumentals are not discussed in Frey & Pittner (1998), Maienborn & Schäfer (2011), Maienborn et al. (2016) and Maienborn (2019). If an affirmative instrumental expresses that an object was acted-on in a given event, the privative instrumental expresses that such an object was not acted-on. Similarly, an affirmative comitative_s states that individuals participated together in an event, while the participation of the referent of the comitative's complement is denied by a privative comitative.

The difference between affirmative and privative comitatives and instrumentals determines their syntactic distribution in German, as has been illustrated in (2), (3), (5), and (6): *affirmative* comitative_s prefer a position to the left of the object in German, but *privative* comitative_s show negligence regarding a position to the left or to the right of the object, while instrumentals prefer a position after the object regardless the affirmative/privative distinction.

Privative comitatives also provide an argument against relating comitative modification to coordination. Consider the privative comitative_s in (11a). Even if it would be possible to provide a coordinated paraphrase, which sounds awkward enough, it *differs* in meaning. As will be discussed below, examples like (11a) do not imply that the referent of the complement of the privative comitative preposition does not act as the subject does. It only implies that they do not participate together in the same event. The only available interpretation for the already rather awkward (11b), however, is that the referent of the subject in the first conjunct mashed garlic, while the referent of the subject in the second conjunct did not.

- (11) a. Ein Hobbykoch hat ohne seinen Assistenten 15 kg Knoblauch
 an.NOM amateur.chef.NOM AUX.SG without his.ACC assistant.ACC 15 kg garlic
 püriert.
 mashed.PTCP
 'An amateur chef mashed 15 kg of garlic without his assistant.'
- b. Ein Hobbykoch und nicht sein Assistent haben 15 kg Knoblauch
 a.NOM amateur.chef.NOM and not his.NOM assistant.NOM AUX.PL 15 kg garlic
 püriert.
 mashed.PTCP
 'An amateur chef, and not his assistant, mashed 15 kg of garlic.'

2.2 The semantic representation of comitatives and instrumentals

Following Rissman & Rawlins (2017), we assume that instrumental prepositions modify a verbal projection denoting an event but do not introduce a dedicated thematic role labelled *instrument*. Instead, Rissman & Rawlins (2017: 509) propose that instrumental interpretations arise because "*an Instrument is an entity ... acted on by an agent as part of a larger event*," where the presence of an agentive thematic role within the broader event is presupposed. Further, Rissman & Rawlins (2017: 529f.) require that "[a]gentive events center around the agent exerting force on a patient", and that the pertinent sub-event relates the agent to the patient via

the instrument. This instrument must be part of the force-dynamic core of the event, which is informally introduced as *FORCE(e)* by Rissman & Rawlins (2017). The force-dynamic core is required in their analysis to distinguish instrumental interpretations of *with* from instrumental interpretations emerging from *to use*, as can be illustrated by the examples in (12) from Rissman & Rawlins (2017: 530):

- (12) Background: Chloe is wearing scuba gear to breathe underwater.
 a. Chloe used scuba gear to clean the floor of the tank.
 b. #Chloe cleaned the floor of the tank with scuba gear.

While (12a) is fine, (12b) is infelicitous because the scuba gear is not part of the force-dynamic core of the event; Chloe does not use it to exert force on the tank. We take over Rissman & Rawlins' (2017: 531) proposal for the denotation of *with* for affirmative instrumentals ((13), notation slightly adapted). It introduces the presupposition that the event must be agentive, i.e. an agent role must be present.

- (13) Instrumental P (affirmative)
 $\lambda x_e. \lambda P. \lambda e_v. \exists y [\text{Agent}(e, y)]. P(e) \wedge \exists e' [e' \subset \text{FORCE}(e) \wedge \text{Agent}(e') = \text{Agent}(e) \wedge \text{ACTED-ON}(e', x)]$

Here, x is the semantic contribution of the complement of the preposition, e the modified event, which is constrained by the semantics of the verb (P). As already pointed out, the instrumental is not treated as a semantic primitive, but instrumental relations are modelled as introducing a subevent e' that is a proper part of the force-dynamic core of the event and is further constrained by the relation *ACTED-ON*.

We will assume that affirmative comitatives introduce a ternary relation that captures concurrence, which will be called *CONCUR*. A meaning representation capturing this is given in (14).

- (14) Comitative P (affirmative)¹¹
 $\lambda x_e. \lambda P_{vt}. \lambda e_v. \exists y [\exists \theta_{(v,e)t} [\theta(e, y) \wedge \theta \in \mathcal{T}(P)] \wedge P(e) \wedge \exists e' [e' \subseteq e \wedge \text{CONCUR}(e', y, x)]]$,
 where for all Q_{vt} :
 $\mathcal{T}(Q) = \{\theta_{(v,e)t} \mid \forall e_v [\Box [Q(e) \rightarrow \exists z [\theta(e, z)]]] \wedge \neg \exists \theta'_{(v,e)t} \forall e'_v \forall z'_e [\theta'(e', z') \rightarrow \theta(e', z')] \wedge \neg [\theta(e', z') \rightarrow \theta'(e', z')] \wedge \forall e''_v [\Box [Q(e'') \rightarrow \exists z'' [\theta'(e'', z'')]]]] \}$

The analysis follows Stolz et al. (2006) in assuming that the ternary relation is to be interpreted so that two individuals are co-present in the same space and time and concur in an event. The comitative requires this event to be a sub-event of the modified event ($e' \subseteq e$). To capture the required relation to a participant of the modified event, we will assume that the non-event argument of the *CONCUR* relation that is not the internal argument of the comitative *P* must be identified with an individual bearing some predicate-specific thematic role in the event introduced by the predicate modified. Formally, we take thematic roles to be relations between events and individuals. Among all of these, the thematic roles occurring with an event of a certain kind (P and Q in (14)) are identified (in the definition of $\mathcal{T}$ in (14), which is to be understood as the set of individual thematic roles required by a predicate of events) as those relations for which there has to be an individual that stands in this relation to the event in question if it is to be an event of this kind ($\forall e_v [\Box [Q(e) \rightarrow \exists z [\theta(e, z)]]]$ in (14)). E.g., in a signing event,

¹¹ $(v,e)t$ stands for the type of functions from ordered pairs of eventualities and entities into truth values here.

there will be someone who signs and something that is signed. But since a signer is an acting individual and someone who writes, there must be an acting individual and someone who writes in a signing event, so this criterion alone does not single out predicate-specific thematic roles. In addition, constraints are necessary to prevent potential ambiguities when considering relations that apply to multiple event participants – for instance, the relation ‘*e contains x as a conscious event participant*’, which holds for both the buyer and the seller in a selling event. To address this, we require that no other relation between events and individuals – mandated by the event type – may entail the target relation, unless the entailment is bidirectional.¹² This ensures that only predicate-specific thematic roles and relations are included, while more general or non-specific ones are excluded (the latter by $\forall e_v[\Box [Q(e) \rightarrow \exists z[\theta(e,z)]]]$ and $\forall e''_v[\Box [Q(e'') \rightarrow \exists z''[\theta'(e'',z'')]]]$). For example, a signer is also a writer, so the role of writer is not predicate-specific for signing and is therefore not included in $\mathcal{T}(\text{sign})$ (technically $\forall e'_v \forall z'_e[[\text{signer}(e',z') \rightarrow \text{writer}(e',z')]]$ holds, so the second conjunct in the condition is not fulfilled and writer is not in $\mathcal{T}(\text{sign})$). A buyer is a conscious event participant so ‘*e contains x as a conscious event participant*’ is not included in $\mathcal{T}(\text{sell})$.

Predicate-specific thematic roles are needed here because the comitative in e.g. a signing event should not only be required to be agentive or a writer, but to sign. It will be crucial for our analysis in section 4, however, that a signer is always a writer, is always someone acting, is always an agent. This is because we will refer to agents in our analysis of constituent linearization. The specification in (14) does not directly identify the thematic role of the preposition’s complement with the one of the target referent: we take this to follow from a meaning postulate on what it means to stand in a concurrence relation, given in (15). This division of labour facilitates a uniform treatment of affirmative and privative comitatives, as we will show below.

$$(15) \quad \forall e_v \forall P_{vt} \forall \theta_{(v,e)t} \forall x_e \forall y_e \forall e'_v [[e' \subseteq e \wedge \text{CONCUR}(e', y, x) \wedge P(e) \wedge \theta(e,y) \wedge \theta \in \mathcal{T}(P)] \rightarrow [\theta(e',x) \wedge P(e')]]$$

The condition (15) states that whenever there is a concurrence relation between two individuals in an event e' , which is a subevent of a larger event e , and the target referent has a certain role within e , the internal argument must bear that role in e' and e' must be the same kind of eventuality as e . E.g., when e is a signing event, there must be a signer and something signed. (14) requires the internal argument of the P to concur with one of them. When the internal argument of the preposition stands in a concurrence relation with the signer, it will stand in the same relation to a subevent of the signing event as the signer stands to the signing event, namely being a signer. Since every signer is an agent, it will also be an agent. When the P ’s internal argument concurs with the signed element on the other hand, it will be a signed element, bearing the role of a patient or theme.

The examples in (3), (4), and (8) illustrate different thematic roles that the comitative can obtain in this way (agents, patients, stimuli). The present analysis departs from previous accounts of comitative prepositions (e.g., Baker 1997; Frey & Pittner 1998; Zhang 2007; Franco & Manzini 2017) by explicitly capturing the flexibility of the comitative relation. The target referent may assume any thematic role that is available within the type of event denoted by the verbal projection modified by the comitative PP. Naturally, (15) permits the comitative to enter into a thematic relation with the event only if a concurrence relation holds. Such a relation is absent in the case of privative prepositions.

¹² $\neg[\theta(e',z') \rightarrow \theta'(e',z')]$ is needed to prevent θ itself and equivalent relations from being excluded.

Privative instrumentals and comitatives differ from affirmative ones in that their meaning contains a negation. As was already pointed out in the previous sections, the difference between affirmative and privative has little consequence for the serialization patterns of instrumentals, but influences the serialization patterns of comitatives, for reasons that will become clear in section 4. For privative instrumentals, introduced by *ohne* ('without') in German, we will assume the following representation:

(16) Privative instrumental

$$\lambda x_e. \lambda P. \lambda e: [\exists y [\text{Agent}(e, y)]] . P(e) \wedge \neg \exists e' [[e' \subset \text{FORCE}(e) \wedge \text{Agent}(e') = \text{Agent}(e)] \wedge \text{ACTED-ON}(e', x)]$$

Consider the case of the privative instrumental in (17):

- (17) Monika hatte ohne einen Schlüssel die Tür geöffnet. Sie
 Monika.NOM AUX.SG without a.ACC key.ACC the.ACC door.ACC open.PTCP she.NOM
 hatte keinen gefunden.
 AUX.SG none.ACC find.PTCP
 'Monika had opened the door without a key. She hadn't found any.'

There is no assertion in the first sentence in (17) that there are *no* keys, but only an assertion that there are no keys that have been acted on by Monika in a subevent e' of e , the opening event. Since (19) does not presuppose that there are no keys, it can be continued as shown. We can thus conclude that privative instrumental PPs neither assert nor presuppose that the restriction of their complement – *Schlüssel* ('key') in (17) – denotes the empty set. Also, it is open in the situation described in (17) whether Monika acted on some other implement to open the door. All of this is consistent with the definition provided in (16). It is equivalent to the formulation in (16'), which states that the agent does not act upon the complement of the preposition in subevents sharing the agent with the main event and forming part of the force-dynamic core of the modified event.

(16') Representation of privative instrumentals (logically equivalent to (16))

$$\lambda x_e. \lambda P. \lambda e: [\exists y [\text{Agent}(e, y)]] . P(e) \wedge \forall e' [[e' \subset \text{FORCE}(e) \wedge \text{Agent}(e') = \text{Agent}(e)] \rightarrow \neg \text{ACTED-ON}(e', x)]$$

As in the case of instrumentals, the difference between affirmative and privative comitatives lies in the presence of a negation in the meaning of the latter: What is denied is the existence of a concurrence event.

(18) Privative comitatives:

$$\lambda x_e. \lambda P_{vt}. \lambda e_v. \exists y [\exists \theta_{(v,e)t} [\theta(e, y) \wedge \theta \in \mathcal{T}(P)] \wedge P(e) \wedge \neg \exists e' [e' \subseteq e \wedge \text{CONCUR}(e', y, x)]]$$

Since the concurrence event is denied ($\neg \exists e' [e' \subseteq e \wedge \text{CONCUR}(e', y, x)]$), (15) cannot apply (because the antecedent of the implication contains $e' \subseteq e \wedge \text{CONCUR}(e', y, x)$) and the internal argument will *not* take over the thematic relations of another participant, even though there *is* a target referent. The existence of a target referent is crucial because concurrence is only denied for one participant. Consider the ambiguous example (19), in which either the concurrence of Clara and Maya or the concurrence of Clara and Anna is denied.

- (19) Maya hatte Anna ohne Clara. verdroschen.
 Maya.NOM AUX.SG Anna.ACC without Clara.ACC beat.up.PTCP
 ‘Maya had beaten up Anna without Clara.’

The presence or absence of specific roles will account for the different serialization patterns of affirmative and privative comitative PPs (see section 4).

3 Experimental Studies

We conducted two experimental studies examining the linear positioning of comitatives and instrumentals in German, using Bayesian Generalized Linear Mixed Models (Bayesian GLMMs, see Vasishth 2023, Franke 2023 and Nicenboim et al. 2025 for introductions aimed at linguists), to analyse the data. We developed prior distributions informed by the accounts of Frey & Pittner (1998), Haider (2000a), and Maienborn et al. (2016). All models were implemented using the *brms* package (Bürkner 2017).

We endeavoured to formalize the core predictions of the theoretical frameworks as prior distributions as faithfully as possible. In some cases, direct translation was not feasible: in the analysis of the first experimental study, Maienborn et al. (2016) could not be operationalized straightforwardly (see fn. 12), and in the analysis of the second study, priors could not be derived directly from Haider (2000a) and Maienborn et al. (2016). Where direct encoding was not possible, we constructed informed or weakly informative priors that approximate the theoretical intent without committing to claims not explicitly made.

Both experimental studies employ objects that are realized as *wh*-indefinites. As was pointed out in the introduction, Frey & Pittner (1998), and Haider (2010) assume existential *wh*-indefinites to be scrambling-invariant: Other elements may scramble above them, but when phrases are found to the right of an *wh*-indefinite, they must have been base-generated there or below (since scrambling is to the left and upwards, respectively). *Wh*-indefinites are, however, ambiguous, also allowing specific interpretations, under which scrambling is no longer assumed to be constrained. We have thus included disambiguating addenda that render specific interpretations contradictory, as is illustrated with test items for instrumentals from experimental study 1 in (20).

- (20) a. Max hat erzählt, dass ein Pfleger über eine Pipette
 Max has told that a.NOM care.assistant.NOM by.means.of a.ACC pipet.ACC
 was verabreicht hat. Was es war, weiß ich aber nicht.
 what.ACC administer.PTCP has what.NOM it was know I but not
 ‘Max said that a care assistant administered something using a pipet. But I don’t know what it was.’
 b. Eva hat erzählt, dass eine Hausfrau ganz ohne ein Spülmittel
 Eva has told that a.NOM housewife.NOM entirely without a.ACC detergent.ACC
 was gereinigt hat. Was es war, weiß ich aber nicht.
 what.ACC clean.PTCP has what.NOM it was know I but not
 ‘Eva said that a housewife cleaned something without using any detergent. But I don’t know what it was.’

3.1 Experimental Study 1 (EXP1)

EXP1 implemented a 2×2 factorial design crossing two types of adverbials (comitatives_s vs. instrumentals) with two types of prepositions showing affirmative or privative senses, respectively. The example pairs provided in (2) (affirmative instrumental), (3) (affirmative comitatives_s), (5) (privative instrumental), and (6) (privative comitatives_s) illustrate the conditions and choices. The design is summarized in table 2, where response option (a) corresponds to PP < OBJ and (b) to OBJ < PP.

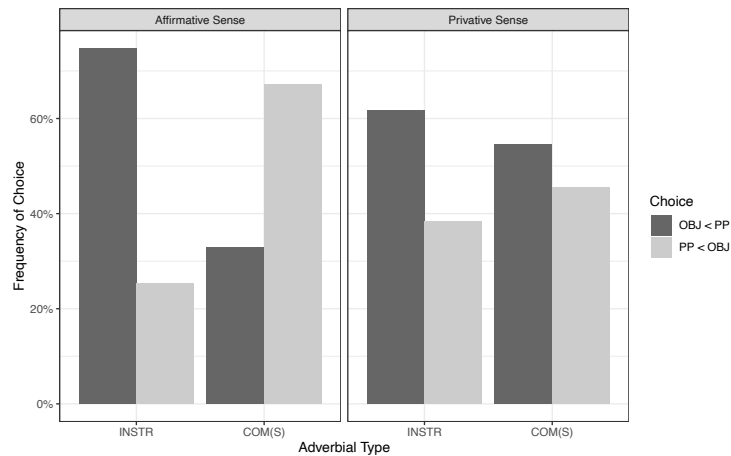
Table 2: 2×2 factorial design in EXP1

Sense		Adverbial Type	
		comitative _s	instrumental
	affirmative	(3a) vs. (3b)	(2a) vs. (2b)
	privative	(6a) vs. (6b)	(5a) vs. (5b)

Each participant evaluated 24 pairs of test items (and 48 pairs of fillers, see the supplementary files provided in <https://osf.io/aqj3z/> for details); 33 participants were included in the final analysis.

Empirical response distributions, as shown in figure 1, indicate a general preference for the serialization OBJ < PP, particularly in instrumental and privative contexts. Only affirmative comitatives_s showed a preference for PP < OBJ. These results suggest that serialization preferences vary by adverbial type and prepositional sense, and that there is a likely an interaction between these two factors.

Figure 1: Empirical Distribution of EXP1



3.1.1 Prior assumptions derived from Frey & Pittner (1998), Haider (2000a), and Maienborn et al. (2016)

Theoretical accounts were encoded as priors over the model terms. For each prior, we have fitted a Bayesian GLMM with *brms* (Bürkner 2017) using *Stan* (Stan Development Team 2023)

in R (R Core Team 2023).¹³ Models included population-level effects for SENSE, TYPE and their interaction, and group-level effects for subjects (slopes) and items (intercepts). Effects were treatment-coded, the reference levels being *affirmative* and *instrumental*. The model formula in brms syntax is: `ANSWER ~ 0 + Intercept + TYPE * SENSE + (0 + TYPE * SENSE | subjects) + (1 | items)`.

Haider (2000a) posits no ordering preference; this is encoded as a symmetric prior on the model intercept (representing affirmative instrumentals) centred at 0 on the logit scale (median probability = 0.5 for PP < OBJ), namely *student's t*(7, 0.0, 0.8). The priors for the treatment-coded effects in the model described below were chosen so that there is practically no difference between the conditions, i.e. the prior for the conditional effect is centred on 0, and not very wide (*student's t*(7, 0.0, 0.6)). The prior based on Haider's predictions can thus count as a neutral one. Frey & Pittner (1998) predict PP < OBJ for affirmative instrumentals; this is encoded as a right-skewed prior (*student's t*(7, 2, 1.4), estimated probability of PP < OBJ $\approx$ 0.88). Maienborn et al. (2016) favour OBJ < PP for instrumentals; which is encoded as a left-skewed prior (*student's t*(7, -2.0, 1.4), estimated probability of PP < OBJ $\approx$ 0.12).¹⁴

Since all three models characterize adverbials at the type level, and do not recognize the distinction between affirmative and privative prepositions, the priors for the other effects were centred at 0 on the logit scale, so that no difference between conditions is assumed. Each prior was defined with wide 95% highest posterior density (HPD) intervals to reflect theoretical uncertainty (after all, the authors make theoretical claims but do not make predictions for the behaviour of participants in the concrete experimental setting we used) and permit substantial posterior updating. The prior distributions are summarized in table 3.

Table 3: Prior predictive distributions for conditional effect of *instrumental*, *affirmative* on percentage scale (share of PP < OBJ predicted) based on priors reflecting Haider's (2000a), Frey & Pittner's (1998), and Maienborn et al.'s (2016) assumptions

	Median Probability	Lower HPD	Upper HPD
Haider (2000a)	0.50	0.13	0.87
Frey & Pittner (1998)	0.88	0.34	1.00
Maienborn et al. (2016)	0.12	0.00	0.66

3.1.2 Models

We are not mainly interested in the marginal posteriors for the population-level effects of the models we fitted, but in the conditional posterior predictions for the four conditions (and some

¹³ This pertains to the main analysis. A sensitivity analysis containing many more models was conducted (and is contained in the OSF) to check in how far the results depend on the exact parameter choice.

¹⁴ As already discussed in fn 6, Maienborn et al. (2016: 328–331) claim that instrumentals (as well as comitatives) mirror the syntactic behaviour of locatives, which would allow them to appear with equal distribution, but with different semantics, before and after objects. While Maienborn et al. (2016: *ibid.*) – as well as Maienborn (2001: 195) – provide evidence that their claims are met by locatives, they do not provide evidence for instrumentals, which do not show a difference in semantics or affect the prosodical structure if they appear in front of the object. If we had ignored the fact that instrumentals do not show a difference in semantics, or do not affect the prosodical structure, but had modelled Maienborn et al.'s assumption accordingly, the prior would have been the same as the one assumed for Haider (2000a).

of the differences between them, see below), which are given in table 4 for the three models with the priors discussed in section 3.1.1.

Table 4: Posterior predictions for the share of PP < OBJ

Prior \ Condition	Haider (2000a)	Frey & Pittner (1998)	Maienborn et al. (2016)
	Posterior Median Probability [lower/upper HPD]		
Affirm. Instrumental	0.29 [0.20, 0.39]	0.28 [0.19, 0.39]	0.27 [0.18, 0.36]
Affirm. Comitative	0.65 [0.54, 0.75]	0.65 [0.54, 0.75]	0.64 [0.53, 0.76]
Priv. Instrumental	0.33 [0.20, 0.46]	0.32 [0.20, 0.46]	0.31 [0.19, 0.44]
Priv. Comitative	0.46 [0.31, 0.62]	0.46 [0.31, 0.61]	0.46 [0.31, 0.62]

They indicate a preference for OBJ < PP for instrumentals (both affirmative and privative), a preference for PP < OBJ for the affirmative comitative condition, and a lack of a preference for the privative comitative condition. This distribution diverges from the predictions of both Frey & Pittner (1998) and Haider (2000a). It aligns more closely with Maienborn et al. (2016) for affirmative instrumentals, but not so for the other conditions.

To test theoretical hypotheses, we computed Bayes Factors (BF) between the models of interest, i.e. the models seeing the experimental data, and null models, i.e. the models where the effect of interest is set to 0. Bayes Factors quantify the relative support for hypotheses. Here, we will report BF_{10} , which reflect the evidence the data provide for the models of interest over the null models. Values greater than 1 speak for the existence of an effect, values smaller than 1 speak against an effect, and in favour of the null model. Since the values are ratios, they are symmetric around 1. Values greater than 3 or smaller than 0.33 are considered moderate, values greater than 10 or smaller than 0.1 are considered strong, and values greater than 100 or smaller than 0.01 are considered decisive (Jeffreys 1939).

Given that the analyses assume that properties are determined at the class level alone, the priors derived from the three proposals mainly differ in the prediction for affirmative instrumentals in relation to fixed objects. Whatever the choice for affirmative instrumentals, it should generalize to the other conditions. In addition to gauging individual hypotheses, we will address the following two questions across all three priors:

- Do we find evidence for the hypothesis that affirmative and privative comitatives show the same choice preferences?
- Do we find evidence for the hypothesis that affirmative instrumentals and affirmative comitatives show the same choice preferences?

The results are summarized in table 5.

Table 5: Hypothesis testing for EXP1 using Bayes Factors and Evidence Ratios. Bayes Factors computed as Savage-Dickey density ratios either on the effects in the model directly or

based on conditional effects (in the latter case using the R packages *emmeans* (Lenth 2023) and *bayestestR* (Makowski et al. 2019))

Hypothesis	BF ₁₀ / Evidence Ratio ¹⁵
(Haider): Equal distribution of serialization with affirmative instrumentals (H ₀ : Intercept = 0)	> 100
(Frey & Pittner): Probability of PP < OBJ higher than OBJ < PP for affirmative instrumentals (H ₀ : Intercept > 0)	< 0.001
(Maienborn et al.): Probability of OBJ < PP higher than PP < OBJ for affirmative instrumentals (H ₀ : Intercept < 0)	> 100
Difference between affirmative and privative comitatives (H₀ for all models: no difference)	
Haider (2000a)	4.76
Frey & Pittner (1998)	4.29
Maienborn et al. (2016)	4.26
Difference between affirmative instrumentals and affirmative comitatives (H₀ for all models: no difference)	
Haider (2000a)	> 100
Frey & Pittner (1998)	> 100
Maienborn et al. (2016)	> 100

The Bayes Factors show decisive evidence in favour of a serialization preference, and hence against Haider’s (2000a) assumption of ordering neutrality. Also, it shows decisive evidence against Frey & Pittner’s (1998) hypothesis that (affirmative) instrumentals prefer a position to the left of the object. Finally, the Bayes Factors speak in favour of Maienborn et al.’s (2016) hypothesis that instrumentals prefer a position to the right of the object (which is in line with the experimental results for instrumentals, but not across all adverbial types tested).

We find moderate evidence against an identical distribution of affirmative and privative comitatives_s, and once again decisive evidence against an identical distribution of affirmative instrumentals and comitatives.¹⁶

3.2 Experimental study 2 (EXP2)

EXP2 extends the investigation by manipulating the object’s syntactic status: objects were realized either as *wh*-indefinites (assumed to be scrambling-invariant in the literature) or as full NP indefinites. Frey & Pittner (1998) predict differential ordering for instrumentals (which should precede the object) and comitatives targeting the object (which should follow the object) if it is fixed. EXP2 tests whether the preference for OBJ < PP with instrumentals in EXP1 can

¹⁵ BF₁₀s are reported for point hypotheses, evidence ratios between the hypothesis and its inverse for one-sided hypotheses, i.e. the ratio of the posterior probability of the hypothesis and the posterior probability of its inverse.

¹⁶ A sensitivity analysis, in which BFs are computed for many priors differing in parameter choice, which is necessary because BFs heavily depend on the prior, shows that evidence against an identical distribution of affirmative and privative comitatives_s is merely anecdotal for models in which the priors for the population-level effects are very narrow, i.e. in which one is very sure a priori that there is no difference between conditions.

be attributed to the possibility of scrambling of the object. In this case, it is expected that OBJ < PP is not found with scrambling invariant objects (and there should be a PP < OBJ preference for comitatives₀). The alternation between scrambling-invariant and flexible objects is illustrated for instrumental PPs in (21) and (22).

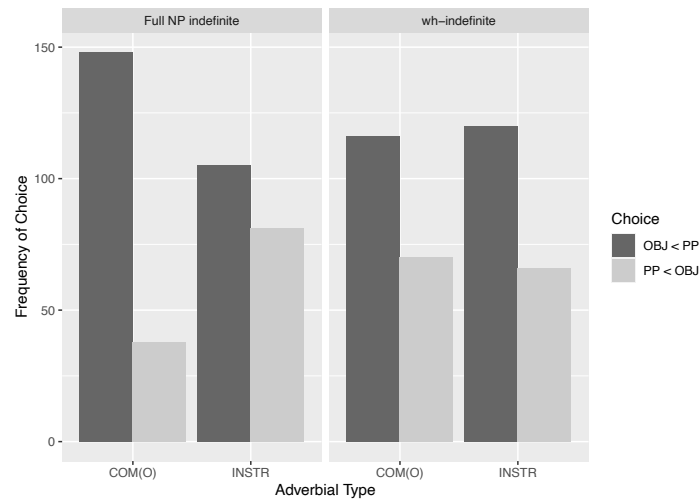
- (21) a. Eva hat erzählt, dass ein Mediziner was mittels einer
 Eva AUX.SG tell.PTCP that a.NOM physician.NOM what.ACC by.means.of a.GEN
 Spritze injiziert hat. Was es war, weiß ich aber nicht.
 syringe.GEN inject.PTCP AUX.SG what.NOM it was know I but not
 b. Eva hat erzählt, dass ein Mediziner mittels einer Spritze was injiziert hat. Was es war,
 weiß ich aber nicht.
 'Eva said that a physician injected something using a syringe. But I don't know what
 it was.'
- (22) a. Eva hat erzählt, dass ein Mediziner eine Substanz mittels
 Eva AUX.SG tell.PTCP that a.NOM physician.NOM a.ACC substance.ACC by.means.of
 einer Spritze injiziert hat. Was für eine es war, weiß ich aber nicht.
 a.GEN syringe.GEN inject.PTCP AUX.SG what.NOM for one it was know I but not
 b. Eva hat erzählt, dass ein Mediziner mittels einer Spritze eine Substanz injiziert hat.
 Was für eine es war, weiß ich aber nicht.
 'Eva said that a physician injected a substance using a syringe. But I don't know which
 one it was.'

Thus, we may adduce evidence against a possible explanation of the serialization pattern OBJ < PP: If the object is not invariant, an analysis relying on scrambling as re-merger to the left might propose that – as an alternative to being base generated – the serialization may result from scrambling the object, as is illustrated in (23b).

- (23) a. [_V OBJ [_V PP ...]]
- b. [_V OBJ_i [_V PP [_V ~~OBJ_i~~ ...]]]

The analysis in (23b), however, cannot be applied to account for the position of scrambling-invariant objects to the left of an event-internal adverbial. If the form of the object does not show a sufficiently strong effect, and PP < OBJ is preferred with instrumental PPs and *wh*-indefinites, then the analysis in (23b) cannot be maintained, which will be translated accordingly into our prior assumptions (see below). Data from 31 participants entered the analysis, who rated 24 minimal pairs each (see the supplementary files provided in <https://osf.io/aqj3z/> for details). Empirical response distributions – as shown in figure 2 – show a preference for OBJ < PP across all conditions.

Figure 2: Empirical Distribution of EXP2



3.2.1 Prior assumptions

We operationalize models with two kinds of priors:

While the predictions of Frey & Pittner (1998) regarding EXP2 can be translated into prior assumptions, this does not extend to the other two proposals for reasons discussed below. As for Frey & Pittner (1998), we can assume the following hypotheses:

- Instrumental PPs should precede the object if the position of the object is fixed (i.e. if the object takes the form of a scrambling-invariant *wh*-indefinite),
- Comitative PPs targeting the object should follow the object if the position of the object is fixed,
- When the position of the object is *not* fixed, instrumental PPs should occur more often after the object (compared to the *wh*-indefinite condition) because the object may scramble, hence the behaviour of instrumental PPs under this condition should differ from their behaviour in the first condition,
- Comitative PPs targeting the object should not be affected from changing the form of the object, because they are already required to appear *after* the object, and scrambling is re-merger to the left.

Similar considerations cannot be derived from the other two proposals, because comitatives₀ are not discussed in these proposals. To provide an alternative prior model to the one derived from Frey & Pittner (1988), we have defined the following weakly-informative prior assumptions, which take their starting point from Haider's assumption that single PPs may occur in any position in German clause structure. We have combined this assumption with the plausible idea that comitatives₀ are more likely to appear after the object:

- Instrumental PPs may appear either before or after the object if the position of the object is fixed,
- Comitative₀ PPs should follow the object if the position of the object is fixed,
- Instrumental PPs should occur more often after the object if the position of the object is *not* fixed,
- Comitative₀ PPs should not be affected by changing the form of the object.

There is thus some correspondence between the two priors, the major difference being in the first condition, which, however, also affects the settings for the other conditions, as is illustrated in table 7. The values in table 7 are prior probabilities of $PP < OBJ$.

Table 7: Prior predictions for conditional effects of prior based Frey & Pittner (1998), and a weakly informative prior

	Weakly informative prior	Frey & Pittner (1998)
	Prior Median Probability [lower/upper HPD]	
Instrumental, fixed object	0.50 [0.13, 0.87]	0.88 [0.14, 1.00]
Comitative, fixed object	0.12 [0.00, 0.98]	0.12 [0.00, 0.97]
Instrumental, variable object	0.27 [0.00, 0.90]	0.62 [0.01, 1.00]
Comitative, variable object	0.12 [0.00, 0.99]	0.12 [0.00, 0.99]

3.2.2 Models

Using these priors, two Bayesian GLMMs were fitted with the following brms formula: $ANSWER \sim 0 + Intercept + TYPE * OBJform + (0 + TYPE * OBJform | subjects) + (1 | items)$ (reference levels of treatment-coded effects: *Instrumental*, *fixed object*). Posterior summaries for the conditional effects suggest little to no difference across instrumental conditions, with a moderate preference for $OBJ < PP$. Comitatives consistently prefer $OBJ < PP$, with even lower probabilities for $PP < OBJ$, as reported in table 8. Just as the priors in table 7, the values in table 8 are reported on the response scale, i.e. as probabilities to realize a PP in front of the object.

Table 8: Posterior summaries for conditional effects of the two models based on a weakly informative prior, and the prior derived from Frey & Pittner (1998)

	Weakly informative prior	Frey & Pittner (1998)
	Posterior Median Probability [lower/upper HPD]	
Instrumental, fixed object	0.38 [0.23, 0.53]	0.39 [0.23, 0.56]
Comitative, fixed object	0.30 [0.14, 0.49]	0.28 [0.12, 0.46]
Instrumental, variable object	0.40 [0.24, 0.56]	0.39 [0.23, 0.56]
Comitative, variable object	0.14 [0.05, 0.26]	0.14 [0.04, 0.26]

As in EXP1, we evaluated hypotheses derived from the theories using Bayes Factors, presented in table 9. The first hypothesis (corresponding to Haider’s view if Haider’s analysis is extended to comitative_o) is that instrumentals can appear before or after fixed objects equally well. According to the Bayes Factor, we neither have evidence in favour nor against this hypothesis.¹⁷ The Bayes Factor for the hypothesis that object form makes a difference is smaller than 0.33, indicating that we find moderate evidence *against* the idea that sentences with a full NP differ from those with a *wh*-indefinite in the environment tested. The same result is obtained with the prior reflecting Frey and Pittner’s (1998) assumptions. Moreover, EXP 2 confirms what

¹⁷ As the sensitivity analysis shows, this holds for priors already assuming that there is no ordering preference. When instrumentals are taken to appear in front of the object a priori, the data provide evidence against it. The more extreme the preference is taken to be, and the surer one is, the stronger the evidence against it.

we have already determined in EXP1: there is decisive evidence against the assumption that instrumental PPs strongly prefer a position before fixed objects.

Table 9: Hypothesis testing for EXP2 using Bayes Factors

Weakly informative prior	BF₁₀ / Evidence Ratio
Instrumentals show equal distribution in relation to fixed objects (H ₀ : Intercept = 0)	1.31
Linearization preferences for instrumentals depend on object type (H ₀ : OBJform = 0)	0.21
Prior derived from Frey & Pittner (1998)	
Linearization preferences for instrumentals depend on object type (H ₀ : OBJform = 0)	0.17
Instrumentals predominantly appear in front of fixed objects (Intercept > logit(65%)) ¹⁸	< 0.01

4 Serialization preferences of comitatives and instrumentals

Capturing the serialization preferences for affirmative and privative comitatives and instrumentals requires a fine-grained analysis, which accounts for the similarities and differences. Haider (2000a: 131) maintains that adjuncts are not individually constrained in terms of linearization; however, he posits that adverbials of different classes must be linearly ordered when combined with a verbal projection. The consequences are similar to the ones proposed in Frey & Pittner (1998): combinations of adverbials with the verb result in configurations that can directly be mapped to serialization. Haider (2000a) does not discuss comitatives or instrumentals, but his analysis implies that their linearization becomes relevant only in combination with other adverbial classes. Thus, the divergent serialization patterns discussed in sections 1 and 3 remain unaccounted for. Maienborn et al. (2016) advance similar assumptions, proposing distinct attachment sites for adverbials and suggesting that comitatives and instrumentals function as V-adjuncts when their relationship to an event participant is semantically licensed. If this assumption is combined with the binary structure of the German clause, serialization patterns again can be derived from configurational conditions. All three analyses propose serialization constraints specifically designed for adverbial modifiers, thereby necessitating distinct serialization rules for arguments and adjuncts.

We consider it preferable to ask whether serialization principles that apply to arguments constrain the serialization of adjuncts as well. Variations in serialization may, in fact, reflect the differential application of well-established linearization constraints. Instrumental modifiers, for example, are consistently realized as prepositional phrases (PPs), are inanimate, and lack agentivity. All these properties are taken to influence the serialization of arguments in the German midfield (see the overview in Frey 2015). In contrast, the animacy of a comitative PP's

¹⁸ The logit is the inverse of the logistic function, which is used in logistic regression models to turn the linear predictor into a probability. The hypotheses tested is that the chance that an instrumental appears in front of a *wh*-indefinite object is at least 65%. The evidence ratio shows that the data provide decisive evidence against this.

complement is influenced by both its target referent and the governing verb. When the referential phrase is an agentive subject, the comitative construction may inherit agentivity. As a result, instrumentals and comitatives respond differently to constraints on agentivity and animacy, while exhibiting parallel behaviour with respect to categorical constraints. Apparent differences between instrumentals and comitatives disappear in the case of privative comitatives, whose internal arguments cannot acquire agentivity. This is because they do not establish a CONCUR relation, and therefore no thematic role is assigned.

We will thus endorse the important insight of Frey & Pittner (1998), Haider (2000a), and Maienborn et al. (2016) that word order constraints should be derived (at least partially) from semantic properties of the adverbials. We do not see the necessity, however, to tie this observation, which, in any case, is incomplete, to the imposition of homogeneous semantic adverbial classes. Instead, we assume that the relevant semantic properties follow from the individual presence or absence of semantic triggers for the linearization constraints. The triggers can be found in adverbials, as well as in arguments of lexical heads, and while some of the triggers are semantic in nature, others are not.

The following sections elaborate on this analysis. Section 4.1 addresses German clause structure and linearization constraints, while section 4.2 presents an account of serialization patterns for comitatives and instrumentals using violable linearization constraints within a Maximum Entropy Grammar framework, and section 4.3 will discuss the impact of the present analysis to proposals that either assume predicate-specific base orders or a dedicated functional projection introducing an external argument.

4.1 On German clause structure, and linear precedence

The present analysis assumes free generation of arguments and adjuncts within the projection of a verbal head. Following Haider (2010), we adopt a binary, right-branching structure for German verbal projections and reject the presence of functional projections between V^0 and C^0 .¹⁹ However, contrary to Haider, we permit any order of combination between arguments and adjuncts, contingent on licensing via categorial or thematic selection or by modification criteria (such as predicate modification, see Heim & Kratzer 1998). We assume no base order and therefore dispense with scrambling as a grammatical mechanism.²⁰ Nonetheless, German clause structure exhibits preferred and dispreferred serializations, necessitating a mechanism to encode these preferences. Traditional *Linear Precedence* (LP) rules, such as those in Gazdar et al. (1985) and Uszkoreit (1987) yield problematic results because their application requires a flat structure of the German clause, which is not tenable. Instead, we will employ a variant of LP rules: linearization constraints that are violable and weighted, thereby accommodating gradient acceptability judgments and preferences among competing serializations, which derive from competing structures. In addition to being violable and weighted, linearization constraints (LCs) differ from LP rules in their domain of application: LP rules apply to local trees, thus imposing the undesirable requirement that all affected constituents must be sisters. LCs, in contrast, will be taken to apply within the maximal projection of a head.

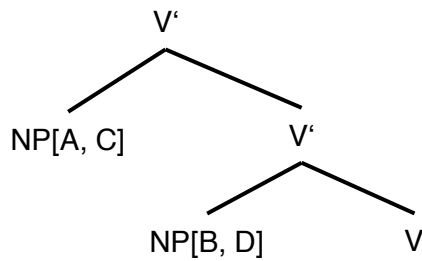
¹⁹ Haider (2010: 58–67) adduces compelling evidence against functional projections between V and C.

²⁰ The long-standing debate whether the observed word order freedom in German should be handled by assuming a fixed base order plus scrambling or free base generation has not yielded a definitive resolution, as has been discussed in Salzmann (2025). The argument that Salzmann (2025) found most compelling, deriving from Heck & Himmelreich's (2017) analysis of intervention with floating quantifiers, has recently been challenged on conceptual and empirical grounds by Kiss et al. (2025).

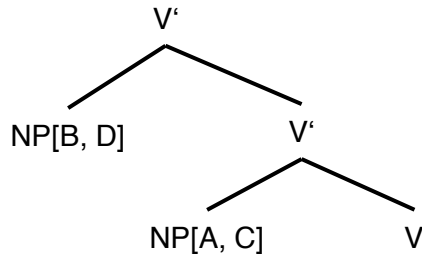
Our analysis draws on *Maximum Entropy Grammar* (MEG; Goldwater & Johnson 2003; Wilson 2006; Hayes 2022), a variant of Optimality Theory (Prince & Smolensky 2004; Müller 2015), where preferences are modelled using weighted constraints. MEG determines a probability for each candidate based on its constraint violation profile. MEG thus also permits multiple candidates to receive the same – or nearly identical – probabilities, thereby accounting for optionality (see Kiss et al. 2025 and the discussion below).

An LC of the form $X < Y$ requires that a phrase having property X found within the projection of a head must be realized to the left of a phrase having property Y within the projection of the same head. As an illustration of the application of LCs, consider the following two tree structures in (24), and the LC $A < B$.

(24) a.



b.



The maximal projections in (24) are generated freely, and (24b) violates $A < B$. If LCs were inviolable, (24b) would be excluded. However, if we assume that linearization constraints (LCs) are violable, then (24a) is merely preferred over (24b).

(25) shows how the probabilities of the candidates are determined based on their constraint violations. Assume that we have m constraints $c_1, \dots, c_m$. For each candidate, constraint violations are counted and weighted: $w_i \times f_i(y, x)$, where f_i is the frequency of violations of constraint c_i , and w_i is the weight of constraint c_i . Our constraint weights will usually be negative. These values are summed up for all m constraints: $\sum_{i=1}^m w_i f_i(y, x)$, which is a penalty score (*Inverted Harmony*). The Inverted Harmony is exponentiated. The higher the absolute value of Inverted Harmony is, the lower will its exponentiated value be (since Inverted Harmony is negative). The probability of an individual candidate is determined by dividing its individual score by the sum of all scores of the candidates (x is a variable for the context in (25), $Y(x)$ the set of candidates).

(25) (Goldwater & Johnson 2003: 114)

$$P(y|x) = \frac{\exp(\sum_{i=1}^m w_i f_i(y, x))}{\sum_{y \in Y(x)} \exp(\sum_{i=1}^m w_i f_i(y, x))}$$

The decisive part of an MEG analysis consists, of course, of the constraints and their weights. These can be computed from the experimental data using logistic regression since an MEG as defined in (25) corresponds to a logistic regression model for our data. This is what we will do in the next section. It should be pointed out here, however, that participants saw only two of the six possible linearizations of subject, object, and PP in the two experiments reported above. We will assume that participants behave as if there were only two candidates when forced to choose between two. While this seems sensible, it is not necessarily innocuous. The constraint weights obtained here should thus be taken with a grain of salt. Another point in question is that the items were not designed to determine constraint weights in the first place, but to test theoretical predictions of other theories. This required that certain factors known to influence linear order in German could not be equalled out within items (such as weight, when one constituent is a PP, the other a *wh*-indefinite). In general, constraint weights must be determined with examples in which all factors not integrated in the analysis themselves are equalled out and by looking into all pertinent constructions, i.e. those constructions to which the constraints may apply. If e.g. an LC requires *agents* to precede *non-agents*, then this condition may not only be evaluated when applying to comitatives, as we will see in section 4.2, but of course also to sentences containing no adverbials, but complements bearing the *agent* role.²¹ In this concrete case, however, comitatives provide a rare chance to disentangle the influence of agentivity from the one of grammatical functions, case, and topicality (which is correlated with subject-hood), as we will argue below.

4.2 Serialization patterns of comitatives and instrumentals

Comitatives and instrumentals differ across several dimensions. Notably, distinctions observed between affirmative and privative comitatives do not extend to instrumentals. Comitatives can bear the feature [+animate], whereas proper instrumental PPs do not. Affirmative comitatives can receive θ -roles by assuming the role of their target referent, including the θ -role *agent*. Instrumental PPs do not introduce an additional θ -role; and even if they did, it would not be that of an agent. Furthermore, affirmative and privative comitatives differ in that the latter do not assume θ -roles, as discussed in Section 2.2. However, comitatives and instrumentals match in their syntactic category, which is PP. The similarities and dissimilarities find their reflexes in three linearization constraints, which apply equally well to arguments and adjuncts, and have not been specifically designed to determine the order of comitatives and instrumentals. The three constraints are given in (26).

- (26) a. Agentivity: [agent] < [-agent] (−1.07 [−1.87, −0.28]; Uszkoreit 1987; Jacobs 1988; Müller 1999; Frey 2015 among others).
 b. Order of categories: NP < PP (−0.86 [−1.2, −0.52]; Gazdar et al. 1985; Müller 1999)
 c. Animacy: [animate] < [inanimate] (−0.67 [−1.32, −0.02]; Müller 1999; Fanselow 2003)

²¹ It should be noted that the present analysis does not provide an exhaustive list of weighted linearization constraints. Although additional constraints may not impact the cases discussed, their inclusion could shift overall weight distributions. Such shifts may alter the absolute strengths of the constraints but are unlikely to affect their relative strengths as applied here.

The reported weights are determined by a Bayesian logistic regression model with rather neutral priors based on the data from the experimental studies with the following formula:²²

$$\text{ANSWER} \sim \theta + \text{agent_constraint} + \text{anim_constraint} + \text{PP_constraint} + \\ (\theta + \text{agent_constraint} + \text{anim_constraint} + \\ \text{PP_constraint} \mid \text{subjects}) + (1 \mid \text{items})$$

Factors are coded by subtracting the constraint violations of the OBJ < PP variant from the ones of the PP < OBJ variant. Adding the group-level constraints for participants has the effect that in principle every participant has their own MEG and there are parameters that measure the difference between the individual MEGs (participants' individual constraint values are assumed to stem from a normal distribution around the population-level value whose scale parameter is estimated), adding a group-level intercept for items, potentially remaining idiosyncrasies of the items can be taken into account.²³

None of the LCs employed has been specifically designed to deal with adjuncts. Instead, we derive word order preferences from constraints applied uniformly to all constituent types.²⁴

4.2.1 Comitatives

The subjects of action verbs in active sentences are agents, and agents are typically *animate*. A syntactic phrase that bears an agent role is thus affected by two of the three linearization constraints introduced. When a comitative is affirmative, its internal argument will take over the thematic role of the target referent by (15). The semantic properties of comitatives as discussed in section 2.2 thus interact with the linearization constraints: when the target referent of a comitative in the projection of an action verb in an active clause is the subject, the comitative will

²² We assume a student's $t(7, 0.0, 1.9)$ distribution for all constraints. More informative priors for constraint weights were not used, as existing estimates are based on data that are not directly comparable (see the remarks at the end of Section 4.1 and below).

²³ Indeed, there is much variation between participants, especially with respect to NP < PP, where the estimated scale parameter of the normal distribution around the population-level effect is 0.97 [0.75, 1.25]. This is huge when compared to the population-level constraint weight (which is on the same scale), and it suggests that either some speakers have a categorical constraint while others don't, or this constraint is not the true decisive factor (but may be correlated with it). For the other two constraints, the values are lower and HDPs wider (0.52 [0.03, 1.21] for *agentivity*, 0.48 [0.02, 1.11] for *animacy*), but still non-negligible. The estimate for the items' group-level intercept is 0.55 [0.37, 0.76], which indicates that there is indeed some variation.

²⁴ A comparison of the constraint weights with those reported by Masloch et al. (2024), who employ the same framework, reveals that the weight assigned to the *Agentivity* constraint in our analysis is notably lower. Masloch et al. (2024) distinguish between *actor* and *causer* constraints, which together sum to a weight of -6. Beyond the caveats noted at the end of Section 4.1, it is important to emphasize that our study addresses the relative order of an object and a PP, whereas their analysis concerns subject vs. object order. They suggest (p. 30) that part of the effect attributed to their animacy constraint (weighted -1.1) may in fact reflect topicality. As established in previous research (e.g., Siewierska 1993), topics generally precede comments, and topicality has been shown to influence linearization preferences in forced-choice experiments in German (Temme & Verhoeven 2016, among others). However, topicality is also correlated with subjecthood in German (Jacobs 1988), which may be the decisive factor underlying the subject-before-object preference with symmetric predicates such as *ähneln* ('to resemble'), as well as the subject-before-PP preference in passives. PPs may therefore provide a more suitable context for isolating the weight of the *agentivity* constraint itself. To our knowledge, no experimental study has systematically disentangled these factors, highlighting a clear avenue for future research.

take over the agent role (we will discuss experiencer-object verbs as a verb-class with non-agentive subjects in section 4.2.3). In the present analysis, linearization constraints affect all phrases that show agentivity and animacy, which include comitatives_s. If the comitative is realized to the left of the object, two of the three constraints in (26) are met, but *Order of categories* is violated (Inverted Harmony: -0.86). The reverse situation emerges if the comitative is realized to the right of the (inanimate) object, resulting in an Inverted Harmony of -1.74 ($-1.07 + -0.67$).²⁵ There is no candidate that satisfies all linearization constraints at the same time. In this situation, the most probable candidate is the one with the smallest absolute Inverted Harmony value, which will be the candidate ordering the animate and agentive comitative targeting the subject in front of the inanimate object. The other candidate is not ruled out, however: How probable it is, depends on the strength of the difference in Inverted Harmony. In this case, it will still have a probability of ca. 29% ($= \exp(-1.74) / (\exp(-1.74) + \exp(-0.86))$).

The crucial difference between affirmative and privative comitatives is that (15) does not apply to the latter, so that they do not take over a θ -role. Privative comitatives_s introduce animate phrases, and hence their serialization is governed by *Animacy*, but not bearing a role, *Agentivity* does not apply to them. The two applicable constraints show similar weights, and hence account for the near optionality of the privative comitatives_s to appear to the left or the right of the object (0.44:0.56). While the optionality is in line with Haider (2000a), on his proposal it remains unaccounted for why privative comitatives show almost optional serializations, while affirmative ones prefer a position to the left of the object.²⁶

Comitatives targeting the object may take over a thematic role if they are affirmative, but this does not really matter for the present analysis, because the role taken over will be the role of the object, so that object and comitative_o are indistinguishable in terms of thematic roles. In addition, the examples were designed to avoid ambiguity. Hence, they contain inanimate objects and inanimate comitatives so that the *Animacy* constraint does not apply either, leaving it to the *Order of categories*, which rates (27a) over (27b) (0.3:0.7).

- (27) a. Ein Antiquitätenhändler hat eine Bronzeskulptur zusammen mit
a.NOM antiques.dealer.NOM AUX.SG a.ACC bronze.sculpture.ACC together with
einem Gemälde verkauft.
a.DAT painting.DAT sell.PTCP
b. Ein Antiquitätenhändler hat zusammen mit einem Gemälde eine Bronzeskulptur
verkauft.
‘An antiques dealer sold a bronze sculpture along with a painting.’

At this point, the question arises as to whether linearization preferences shift when inanimate phrases are replaced by animate ones. Indeed, the constraints in question predict a corresponding change in preference. We will address these changes following our application of the constraints to instrumentals, as analogous effects are observed for both instrumentals and comitatives that target the object.

²⁵ The comparative influence of animacy will be further discussed in section 4.2.2. For expository purposes, we will ignore the remaining uncertainty about the constraint weights reflected in the full marginal posterior distributions and use only the estimates for computations.

²⁶ Once again, we would like to emphasize that Haider (2000a) does not deal with comitatives or instrumentals but generally claims word order freedom for single occurrences of adverbials in German clauses.

4.2.2 Instrumentals

Instrumentals behave approximately like comitatives_o in preferring a position to the right of the object. This observation stands in contrast to Frey & Pittner's (1998) assumption that instrumentals should behave like comitatives_s. The *Agentivity* constraint does not apply to instrumentals, and the experimental data have been designed to contain inanimate objects only for the examples containing instrumental PPs. Without a difference in *Animacy*, *Order of categories* is the only applicable constraint, and this constraint places PPs to the right of NPs. As with comitatives_o, we observe that this constraint is not always met in the experimental data, but the preference is strong and reflected in the relative weight of the constraint.

The constraints can of course be applied to data that have not come from the experimental studies. To this end, let us consider the effect of applying the *Animacy* constraint if we change inanimate to animate objects in examples with instrumental adverbials.

- (28) a. Da hat ein Verbrecher mit einem Bolzenschneider einen Studenten
 there AUX.SG a.NOM criminal.NOM with a.DAT bolt.cutter.DAT a.ACC student.ACC
 erschlagen.
 slay.PTCP
 b. Da hat ein Verbrecher einen Studenten mit einem Bolzenschneider erschlagen.
 'A criminal slayed a student with a bolt cutter.'

The object in (28) is animate, and hence two of the three constraints are applicable. For instrumental PPs and inanimate objects, the probability of a serialization of the PP to the left of the object is approximately 30 % (which yields a probability of 70 % for a realization to the right of the object). For instrumental PPs and animate objects, as observed in (28), the probability of a serialization of the PP to the left of the *animate* object drops to 18 % ($= \exp(-0.86-0.67) / (\exp(-0.86-0.67) + \exp(0))$). The same considerations apply to comitatives_o, as can be illustrated with (29), a variation of (9).

- (29) a. Er hat einen Schüler zusammen mit einigen Sachen abgeholt.
 he AUX.SG a.ACC pupil.ACC together with some.DAT things.DAT pick.up.PTCP
 b. Er hat zusammen mit einigen Sachen einen Schüler abgeholt.
 he AUX.SG together with some.DAT things.DAT a.ACC pupil.ACC pick.up.PTCP
 'He picked up a pupil along with some things.'

The object of the comitative is not required to share the animacy value with its target referent, so a combination of an animate object, and an inanimate comitative_o, as in (29), is not blocked (counter Zhang 2007: 136). We predict a preference for the serialization in (29a), just as is the case with instrumentals illustrated in (28).

The question arises as to why a serialization in which an instrumental PP or comitative_o PP precedes the object is possible at all, given that the alternative order does not violate any constraints. The straightforward answer is that participants in the experimental study selected this order, which happens to coincide with the sole order for instrumental PPs proposed by Frey & Pittner (1998). We must therefore acknowledge that this order is not ungrammatical – a fact reflected in the nature of the constraints themselves: they are violable and weighted, thereby capturing preferences among alternatives rather than categorical distinctions.

Furthermore, it is important to consider that the semantic relationships between arguments and adjuncts can be readily computed during sentence processing. Once an instrumental is identified as such, it can only target the agent, thereby eliminating any potential ambiguity in

examples (2), (6), and (28). A similar rationale applies to comitative_O constructions: since the complement of the comitative is inanimate, it can be associated with an animate object, but not with an animate subject when the subject bears an agent role, as this role is incompatible with an inanimate interpretation.

4.2.3 Application to experiencer-object verbs

We have already emphasized that comitatives may combine with a larger set of verb classes than instrumentals, because they do not require the presupposition of an agent. This was already illustrated with psychological predicates in (8) and can be further illustrated with experiencer-object verbs (see Rozwadowska 2017 for a survey and Masloch et al. 2024 for a recent analysis on the linearization of arguments of experiencer-object verbs in German). As the name says, the object bears the role of an experiencer, while we will call the subject a stimulus here. We will look at comitatives_S with experiencer-object verbs, as illustrated in (30).²⁷

- (30) a. Ich glaube, dass der Eiswagen dem Mann zusammen mit
 I believe that the.NOM ice.cream.cart.NOM the.DAT man.DAT together with
 den 30 Eissorten gefiel.
 the.DAT.PL 30 ice.cream.flavour.DAT.PL please.PST
- b. Ich glaube, dass der Eiswagen zusammen mit den 30
 I believe that the.NOM ice.cream.cart.NOM together with the.DAT.PL 30
 Eissorten dem Mann gefiel.
 ice.cream.flavour.DAT.PL the.DAT man.DAT please.PST
- c. Ich glaube, dass dem Mann der Eiswagen zusammen mit
 I believe that the.DAT man.DAT the.NOM ice.cream.cart.NOM together with
 den 30 Eissorten gefiel.
 the.DAT.PL 30 ice.cream.flavour.DAT.PL please.PST
- d. Ich glaube, dass dem Mann zusammen mit den 30
 I believe that the.DAT man.DAT together with the.DAT.PL 30
 Eissorten der Eiswagen gefiel.
 ice.cream.flavour.DAT.PL the.NOM ice.cream.cart.NOM please.PST
- e. Ich glaube, dass zusammen mit den 30 Eissorten der
 I believe that together with the.DAT.PL 30 ice.cream.flavour.DAT.PL the.NOM
 Eiswagen dem Mann gefiel.
 ice.cream.cart.NOM the.DAT man.DAT please.PST
- f. Ich glaube, dass zusammen mit den 30 Eissorten dem
 I believe that together with the.DAT.PL 30 ice.cream.flavour.DAT.PL the.DAT
 Mann der Eiswagen gefiel.
 man.DAT the.NOM ice.cream.cart.NOM please.PST
 'I believe that the ice cream cart pleased the man along with the 30 ice cream flavours.'

In (30), the subject is the inanimate stimulus der *Eiswagen* ('the ice-cream van'). A modification targeting the subject does not require the comitative and its target referent to share the same

²⁷ The parallelism between "subject-oriented" comitatives and instrumentals asserted in Frey & Pittner (1998) is further invalidated if experiencer-object verbs are examined. Here, the subject denotes a stimulus, and its animacy value is not constrained by the verb, in contrast to the animacy value of the sentient object. Consequently, the combination of an instrumental with a non-agentive experiencer-object verb is blocked.

animacy value, as experiencer-object verbs generally permit both animate and inanimate stimuli. However, potential ambiguity in the interpretation of the comitative is precluded: targeting the object would necessitate that the comitative’s complement and the object match in animacy. Thus, the possibility of the comitative targeting the experiencer in (30) is blocked in the same way as targeting an animate subject with an inanimate comitative complement in the context of an action verb. Following Masloch et al. (2024), we assume that, with verbs such as *gefallen* (‘to appeal to’), the experiencer may precede or follow the stimulus; however, animacy exerts an influence on serialization, resulting in the six possible orders observed in (30), with their respective probabilities presented in table 10.²⁸

Table 10: Example computation probabilities experiencer-object verb

	Agentivity	NP < PP	Animacy	In. Harm.	exp(In. Harm.)	P
	−1.07	−0.86	−0.67			
(30a)			1	−0.67	0.51	0.23
(30b)		1	2	−2.2	0.11	0.05
(30c)				0	1	0.46
(30d)		1		−0.86	0.42	0.19
(30e)		2	2	−3.06	0.047	0.02
(30f)		2	1	−2.39	0.09	0.04

The application of the linearization constraints results in a ranking that conforms to our intuitions. We notice a clear contrast between (30c) and the other options, which is reflected in the highest probability of realization in table 10. Finally, consider the examples in (31) which differ from (30) in being genuinely ambiguous.

- (31) a. Ich glaube, dass der Eisverkäufer dem Jungen zusammen mit seiner
I believe that the.NOM iceman.NOM the.DAT boy.DAT together with his.DAT
Freundin gefiel.
girlfriend.DAT please.PST
b. Ich glaube, dass dem Jungen der Eisverkäufer zusammen mit seiner Freundin gefiel.
‘I believe the that the iceman pleased the boy together with his girlfriend.’

Adopting the analysis of Masloch et al. (2024), we assume that the order of arguments in (31) does not exhibit a preference; both serializations are equally acceptable, as neither violates any relevant constraint. The comitative, featuring an animate complement, may associate with either the subject or the object in (31), resulting in two distinct interpretations (stimulus or experiencer). However, any alternative placement of the comitative – such as inserting it between the arguments – would incur violations of the *Order of Categories* constraint. Consequently, the analysis predicts that both argument orders in (31) are equally preferred, whereas configurations in which the comitative intervenes between the arguments are dispreferred.

²⁸ A fully-fledged analysis will require additional constraints (e.g. for information structure and weight). One could also include the uncertainty about the constraint weights and possible variation between speakers, but for the expository purposes here, table 10 suffices.

4.3 Thematic relations, functional heads, and (no) fixed base orders for adjuncts and arguments

The present proposal specifically combines free base generation of arguments with the abandonment of functional projections between C and V. This combination of assumptions contrasts with two prominent alternatives. Haider (2000b, 2017) argues in favour of predicate-specific fixed base orders that reflect conceptual structure (and thus thematic notions in a broad sense), while Fanselow (2003) assumes a functional projection introducing the subject, which accounts for the preference for subjects to appear before objects due to their realization in different syntactic domains. A comparison of the present proposal with these alternatives serves to elucidate the plausibility of the competing approaches.

Haider (2000b, 2017) assumes that the subject of an action verb precedes the object in the base order because this order reflects the lexical argument structure, which is itself derived from conceptual structure. The present proposal similarly assumes that the subject of an action verb regularly precedes the object because the former is an agent, and there is a general – albeit violable – principle that requires agents to precede non-agents, viz. *Agentivity*. If only arguments are considered, one might argue that Haider (2000b, 2017) and the present proposal yield similar results.²⁹ However, a crucial feature of the present analysis is that the constraints applicable to arguments are also extended to modifiers – comitatives in particular – thereby accounting for their serialization as well.

The present analysis accounts for the differences between comitative_O and comitatives_S, as well as for the distinction between affirmative and privative comitatives_S, by applying *Agentivity* to these PPs. A comitative PP becomes agentive if the subject is agentive. It is not only the same principle that explains the serialization preferences of subjects and comitative PPs targeting an agentive subject; this principle is also independently necessary to account for the variation among different types of comitatives. Under Haider’s (2000b, 2017) proposal, by contrast, the serialization preferences of subjects and comitatives targeting an agentive subject remain two separate issues – the latter not being accounted for at all.

Similar considerations emerge if we compare the present analysis to an analysis employing a functional projection to introduce the *agent* role. The general idea to introduce the subject as a specifier of *n* has been proposed by Kratzer (1996) and has been taken up specifically to deal with the serialization of subjects in German by Fanselow (2003). Fanselow (2003) uses a dedicated functional head *v* to deal with word order regularities, the specifier of which hosts the agent. The alternative is to assign the *agent* role by the verb, as in the present analysis.

Once again, decisive arguments in favour of either position are hard to find if one concentrates on either the serialization of arguments, or on an analysis to account for differences between comitatives targeting subjects vs. comitatives targeting objects. With its extension to dealing with arguments and adjuncts in a uniform analysis, the present proposal provides a probe into this debate. While proponents of an assignment by a functional head could argue that the serialization of subjects in front of the object follows from subjects and objects being realized in different syntactic domains, it remains unclear under such a view why affirmative comitatives_S should share the preference to be realized to the left of the object with the (agentive) subject. Various proposals have employed a dedicated functional head – either *Voice* or *v* – to account for comitatives in English and other languages, such as Bruening 2013, Franco &

²⁹ We should point out that the similarity of results already comes to an end, if other verb classes than action verbs are considered. Masloch et al. (2024) show that *Animacy* further influences the argument serialization preferences of experiencer-object verb. Since these lack an agent, the preferences are unexpected under Haider’s assumptions.

Manzini 2017, and Chen 2024. They assume that comitatives_S and comitatives_O modify different nodes: a projection of the subject-introducing head in the case of comitatives_S, a lower node in the case of comitatives_O. The semantics of the comitative will then lead to the correct target referent choice. Interestingly, the analyses imply that the comitative will be minimally c-commanded by its target referent unless there are more modifiers of the same head.³⁰ Thus, the analyses run into the same problems as Frey & Pittner (1998) if one were to apply these approaches to German naively. In addition, the proposals are not able to account for the differences in preferences between comitatives and instrumentals – under the proviso that they can be applied to serialization in German. And even if one were willing to assume movement to higher projections in the clause (above VoiceP or nP, respectively) to get the orders right, one would still need an explanation for the orders observed: Why should it be the case that an object scrambles above a comitatives_S and the subject scrambles above it so that SUBJ < OBJ < PP is observed rather frequently with privative comitatives_S, but less so with affirmative ones? It appears that constraints similar to the ones proposed here would be needed in addition. If the serialization properties of agentive subjects and comitatives are to be covered by functional projections, one alternative could be to introduce an additional functional projection for the sole purpose of ordering a subset of comitative PPs.

Of course, the present analysis does not require such a projection. The linearization constraints employed are required anyway, as well as an implementation of the observation that comitatives_S can act as ‘secondary’ agents, which has been discussed in Lakoff & Peters (1966), Baker (1997), Frey & Pittner (1998), Stolz et al. (2006), and Zhang (2007). For the purposes of serialization, it does not matter whether agentivity comes about because of a lexical assignment (by the verb to its subject), or by an identification, as proposed in (17). Both cases result in phrases bearing a role which is taken up by the same linearization constraint, thus accounting for the position of an affirmative comitatives_S in front of the non-agentive, inanimate object.

5. Conclusion

We have provided an analysis of the serialization preferences of comitatives and instrumentals in German clause structure. While both adverbial modifier types share the property of not only modifying an event, but also a participant within the event, the adverbial types differ in many other respects, which is reflected in their serialization preferences. The identification of a target referent is fixed for instrumentals, but flexible for comitatives, and affirmative comitatives differ from privative ones in taking over a θ -role.

The differing properties affect the serialization options of comitatives and instrumentals in different ways, because the uniform linearization constraints imposed here do not look into the difference between arguments and adjuncts, but into properties common to both, such as *agentivity* and *animacy*.

The present analysis thus argues against a categorial constraints. Instead, we have employed weighted, violable constraints in a MEG, which account for preferences, as well as (near-) optionality.

The extension of linearization constraints imposed for arguments to modifiers, as well as the implementation of the targeting mechanism of comitatives in their semantics allows us to

³⁰ It is not entirely clear whether our criticism applies to Franco & Manzini (2017). In discussing their example (33), they seem to indicate that the adverbial merges after the target referent.

provide a probe for comparing uniform linearization constraints with analyses employing predicate-specific based orders or dedicated functional projections for external arguments.

Data availability/Supplementary files

All analysis scripts, the raw data (items and results), and the scripts containing the sensitivity analyses are available from <https://osf.io/aqj3z/>.

Acknowledgements

This paper has benefitted from discussions with Markus Egg, Bart Geurts, Hubert Haider, Joachim Jacobs, Claudia Maienborn, and Gereon Müller. We wish to thank Maria Kouneli and IGRA Leipzig to give us the opportunity to present an earlier version of the analysis at Leipzig University, and the audience for their comments and suggestions. Special thanks go to Jutta Pieper, who has set up and programmed the experimental environment.

References

- Baker, Mark (1997). Thematic roles and syntactic structure. In Haegeman, Liliane (ed.), *Elements of grammar: Handbook of generative syntax*, pp. 73–137. Dordrecht: Kluwer. https://doi.org/10.1007/978-94-011-5420-8_2
- Bürkner, Paul-Christian (2017). brms: An R Package for Bayesian Multilevel Models Using Stan. *Journal of Statistical Software* 80(1). 1–28. DOI: 10.18637/jss.v080.i01
- Bruening, Benjamin (2013). By Phrases in Passives and Nominals. *Syntax* 16(1). 1–41. DOI: 10.1111/j.1467-9612.2012.00171.x
- Chen, Mingjiang (2024). A decomposition analysis of Agents. *Glossa* 9. 1–42. DOI: 10.16995/glossa.10624
- Diesing, Molly (1992). *Indefinites*. Cambridge; Mass.: MIT Press.
- Fanselow, Gisbert (2003). Free constituent order. A minimalist interface account. *Folia Linguistica* 37(1/2): 191–232. <https://doi.org/10.1515/flin.2003.37.1-2.191>
- Franco, Ludovico & Manzini, M. Rita (2017). Instrumental prepositions and case: Contexts of occurrence and alternations with datives. *Glossa* 2(1). 1–37. DOI: 10.5334/gjgl.111
- Franke, Michael (2023). *An Introduction to Data Analysis*. <https://michael-franke.github.io/intro-data-analysis/>
- Frey, Werner (2015). Word order. In: Kiss, Tibor & Alexiadou, Artemis (eds.), *Syntax – Theory and analysis: An international handbook*, vol. 1, pp. 514–562. Berlin/New York: Mouton de Gruyter. <https://doi.org/10.1515/9783110377408>

Frey, Werner & Karin Pittner (1998). Zur Positionierung der Adverbiale im deutschen Mittelfeld. *Linguistische Berichte* 176: 489–534.

Gazdar, Gerald & Klein, Ewan & Pullum, Geoffrey K. & Sag, Ivan A. (1985). *Generalized Phrase Structure Grammar*. Cambridge, Massachusetts: Harvard University Press.

Goldwater Sharon & Johnson, Mark. (2003). Learning OT constraint rankings using a Maximum Entropy model. In Spenader, Jennifer & Eriksson, Anders & Dahl, Östen (eds.), *Proceedings of the Stockholm workshop on 'Variation Within Optimality Theory'*, pp. 111–120. Stockholm: Stockholm Univ.

Haider, Hubert (2000a). Adverb placement: Convergence of structure and licensing. *Theoretical Linguistics* 26: 95–134. <https://doi.org/10.1515/thli.2000.26.1-2.95>

Haider, Hubert (2000b). Branching and Discharge. In Peter Coopmans & Martin Everaert & Jane Grimshaw (eds), *Specification and Insertion*, 135–164. Amsterdam: Benjamins.

Haider, Hubert (2010). *The syntax of German*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9780511845314>

Haider, Hubert (2017). Mittelfeld phenomena: Scrambling in Germanic. In *The Wiley Blackwell companion to syntax, second edition*, 2573–2645. John Wiley & Sons, Ltd. DOI: 10.1002/9781118358733.wbsyncom048

Hayes, Bruce. (2022). Deriving the wug-shaped curve: A criterion for assessing formal theories of linguistic variation. *Annual Review of Linguistics* 8(1): 473–494.

Heck, Fabian & Anke Himmelreich (2017). Opaque Intervention. *Linguistic Inquiry* 48(1): 47–97. https://doi.org/10.1162/ling_a_00235

Heim, Irene & Angelika Kratzer (1998). *Semantics in generative grammar*. Malden, MA: Blackwell.

Jacobs, Joachim (1988). Probleme der freien Wortstellung im Deutschen. In: *Arbeitsbericht Nr. 5 des Forschungsprogramms Sprache und Pragmatik*, 8–37. Lund.

Jeffreys, Harold (1939). *Theory of probability*. Oxford: Clarendon Press.

Kratzer, Angelika (1996). Severing the External Argument from its Verb. In: Johan Rooryck & Laurie Zaring (eds). *Phrase Structure and the Lexicon*. Studies in Natural Language and Linguistic Theory, vol 33. Springer, Dordrecht. https://doi.org/10.1007/978-94-015-8617-7_5

Lakoff, George & Stanley Peters. (1966). Phrasal conjunction and symmetric predicates. *Mathematical Linguistics and Automatic Translation, Report no. NSF-17*, pp. VI/1–49. Cambridge, MA: Harvard University. [Reprinted in 1969 in: David A. Reibel & Sanford A. Schane (eds). *Modern studies in English*. Englewood Cliffs, pp. 113–142. N.J.: Prentice-Hall.]

Lechner, Winfried (1998). Two kinds of reconstruction. *Studia Linguistica* 52(3): 276–310.

Lenth, Russell V. (2023). *emmeans: Estimated marginal means, aka least-squares means (manual)*. <https://CRAN.R-project.org/package=emmeans>

Maienborn, Claudia (2019). Events and states. In: Robert Truswell (ed.), *Handbook of Event Structure*, pp. 24–65. Oxford University Press.

Maienborn, Claudia (2007). On Davidsonian and Kimian States. In: I. Comorovski & K. von Heusinger (eds.), *Existence: Syntax and Semantics*. Dordrecht: Kluwer, pp. 107–130.

Maienborn, Claudia (2005). On the limits of the Davidsonian approach: The case of copula sentences. *Theoretical Linguistics* 31(3): 275–316.

Maienborn, Claudia (2001). On the position and interpretation of locative modifiers. *Natural Language Semantics* 9: 191–240. <https://doi.org/10.1023/A:1012405607146>

Maienborn, Claudia (2000). Zustände - Stadien - stativische Ausdrücke: Zur Semantik und Pragmatik von Kopula-Prädikativ-Konstruktionen. *Linguistische Berichte* 183: 271–307.

Maienborn, Claudia (1995). Towards a compositional semantics for locative modifiers. In: M. Simons & T. Galloway (eds.), *Proceedings from Semantics and Linguistic Theory (SALT) V*, pp. 237–254. Ithaca N.Y.: Cornell University Linguistic Publications.

Maienborn, Claudia, Helga Gese & Britta Stolterfoht (2016). Adverbial modifiers in adjectival passives. *Journal of Semantics* 33(2): 299–358. <https://doi.org/10.1093/jos/ffv004>

Maienborn, Claudia & Schäfer (2011). Adverbs and adverbials. In: von Heusinger, Klaus & Claudia Maienborn & Paul Portner (eds.), *Semantics. An international handbook of natural language meaning; Volume 2*. (HSK 33.2), pp. 1390–1420, Berlin, New York: Mouton de Gruyter.

Makowski, Dominique & Mattan S. Ben-Shachar & Daniel Lüdecke (2019). bayestestR: Describing effects and their uncertainty, existence and significance within the Bayesian framework. *Journal of Open Source Software* 4(40): 1541. DOI: 10.21105/joss.01541

Masloch, Simon, Johanna M. Poppek & Tibor Kiss (2024). Not so peculiar after all: On the normal position of arguments of German experiencer-object verbs, *Glossa: a journal of general linguistics* 9(1). <https://doi.org/10.16995/glossa.10150>

Müller, Gereon (2015). Optimality-theoretic syntax. In Kiss, Tibor & Alexiadou, Artemis (eds.), *Syntax – Theory and analysis: An international handbook*, vol. 2., pp. 875–936. Berlin/New York: Mouton de Gruyter. <https://doi.org/10.1515/9783110363708-003>

Müller, Gereon (1999). Optimality, markedness, and word order in German. *Linguistics* 3: 777–818. <https://doi.org/10.1515/ling.37.5.777>

Neeleman Ad & Tanya Reinhart (1998). Scrambling and the PF interface. In: Butt, Miriam & Geuder, Wilhelm (eds.), *The projection of arguments: Lexical and compositional factors*, pp. 309–353. Stanford: CSLI.

Nicenboim, Bruno & Daniel J. Schad & Shravan Vasishth (2025). *Introduction to Bayesian Data Analysis for Cognitive Science*. <https://bruno.nicenboim.me/bayescogsci/>

Prince Alan & Paul Smolensky (2004). Optimality Theory: Constraint interaction in Generative Grammar. In McCarthy, John (ed.), *Optimality Theory in phonology. A reader*, pp. 1–71. Oxford, UK: Blackwell. <https://doi.org/10.1002/9780470756171.ch1>

R Core Team (2023). *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. <https://www.R-project.org/>

Rozwadowska, Bożena. (2017). Psychological verbs and psychological adjectives. In Everaert, Martin & van Riemsdijk, Henk (eds.), *The Wiley Blackwell Companion to Syntax*, pp. 3383–3408. John Wiley & Sons. <http://doi.org/10.1002/9781118358733.wbsyncom040>

Rissman, Lilia & Kyle Rawlings (2017). Ingredients of instrumental meaning. *Journal of Semantics* 34(3): 507–537. <https://doi.org/10.1093/jos/ffx003>

Salzmann, Martin (2023). *Word order in the German middle field – scrambling*. Retrieved from <https://ling.auf.net/lingbuzz/006866>

Schröder, Jochen (1986). *Lexikon deutscher Präpositionen*. Leipzig: VEB Verlag Enzyklopädie.

Stan Development Team (2022). *Stan Modeling Language Users Guide and Reference Manual*, version 2.33. <https://mc-stan.org>

Siewierska, Anna (1993). On the Interplay of Factors in the Determination of Word Order. In Jacobs, Joachim & von Stechow, Arnim & Sternefeld, Wolfgang & Vennemann, Theo (eds.), *Syntax: An International Handbook of Contemporary Research*, Vol. 1, 826–846. De Gruyter Mouton. DOI: 10.1515/9783110095869.1.13.826

Stolz, Thomas (1998). UND, MIT und/oder UND/MIT? - Koordination, Instrumental und Komitativ - kymrisch, typologisch und universell. *STUF - Language Typology and Universals* 51(2): 107–130.

Stolz, Thomas, Cornelia Stroh & Aina Urdze (2006). *On comitatives and related categories. A typological study with special focus on the languages of Europe*. Berlin/New York: Mouton de Gruyter.

Struckmeier, Volker (2014). *Scrambling ohne Informationsstruktur?: Prosodische, semantische und syntaktische Faktoren der deutschen Wortstellung*. Berlin: Akademie Verlag. <https://doi.org/10.1524/9783110347715>

Temme, Anne & Verhoeven, Elisabeth (2016). Verb class, case, and order: A crosslinguistic experiment on non-nominative experiencers. *Linguistics* 54(4). 769–813. DOI: 10.1515/ling-2016-0018

Uszkoreit, Hans (1987). *Word order and constituent structure in German*. Stanford: CSLI Publications.

Vasishth, Shravan (2023). Some Right Ways to Analyze (Psycho)Linguistic Data. *Annual Review of Linguistics* 9: 273–291. DOI: 10.1146/annurev-linguistics-031220-010345

Wilson, Colin (2006). Learning phonology with substantive bias: an experimental and computational investigation of velar palatalization. *Cognitive Sciences* 30: 945–982. https://doi.org/10.1207/s15516709cog0000_89

Zhang, Niina Ning (2007). The syntax of the English comitative construction. *Folia Linguistica* 42(1-2): 135–169. <https://doi.org/10.1515/flin.41.1-2.135>